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Three-dimensional stress rotation and control mechanism of deep tunneling incorporating generalized Zhang–Zhu strength-based forward analysis

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ABSTRACT

Deep rock tunneling exhibits a significant three-dimensional (3D) space effect. The complex stress path and extrusion deformation during excavation are the most significant and decisive factors in the stability and construction safety of a deep tunnel. 3D and forward numerical analyses based on generalized Zhang–Zhu (GZZ) strength criterion are performed to investigate the principal stress rotation behavior and active control mechanism during deep tunneling. The surrounding rock element experiences significant stress rotations because of the sharply increased shear stress (τ_{ry}) near the tunnel face. A dimensionless stress index (τ_{ry}/I_1) is proposed to quantitatively evaluate the principal stress rotation during excavation. τ_{ry}/I_1 increases with increasing buried depth, and the deep tunnel presents significant principal stress rotation that reaches a large τ_{ry}/I_1 value. The mechanical behavior of the advanced core rock is primarily affected by the strengthening parameter, geological strength index (GSI). Strengthening of the core rock can greatly improve the stress conditions and mechanical behavior, increase the strength of the rock mass, and reduce the stress rotation. The pre-extrusion deformation of the core rock and pre-convergence of the surrounding rock are discovered to follow a consistency law, which suggests that the pre-convergence deformation and surrounding rock stability near the tunnel face depend on the extrusion deformation of the tunnel face. The GZZ strength-based 3D forward analysis and stress control method proposed in this study can enhance the design and construction of deep and ultra-deep tunnels by weakening the 3D space effect (e.g., tunnel face extrusion and stress rotation) and exerting the strength potential (self-bearing capacity) of a rock mass.

1. Introduction

The mechanical behavior of deep rock tunnels under high ground stress and strong excavation disturbance is different from that of shallow tunnels (Yan et al., 2015; Qian and Zhou, 2018; Wagner, 2019). The plane strain model mainly focuses on the radial stress ($\sigma_r = \sigma_3$), tangential stress ($\sigma_\theta = \sigma_1$), and radial displacement (crown settlement and horizontal convergence deformation) of surrounding rocks (Eberhardt, 2001); however, it neglects the 3D space effect near the tunnel face, e.g., complex stress path and extrusion deformation of tunnel face. Therefore, the plane strain model applies to the stability design of the completed tunnel or tunnel section located far away from the tunnel face (Martin, 1997; Eberhardt, 2001; Diederichs et al., 2004), but does not apply to deep rock tunnels that show significant 3D space effects during tunneling.

Deep tunnel disasters, such as rock burst, large deformation, and steel arch distortion occur primarily near the tunnel face. The instability of advanced core rock mass often leads to the collapse of the surrounding rock, but not before the core rock collapses (Lunardi, 2000, 2008; Cai et al., 2022a). However, when the plane strain model is considered, the projected engineering disasters are almost zero. Lunardi (2000) and Hoek et al. (Hoek and Marinos, 2000; Hoek, 2001) argue that the use of plane strain theory for tunnel design should be limited to conditions where the convergent strain in the unsupported surrounding rock is <2.5%. However, when the convergent strain is higher than 2.5%, the 3D space effect of the excavation face impacts the overall stability of the tunnel face, which further influences the overall stability of surrounding rocks (Lunardi, 2000). The plane strain model overemphasizes the radial convergence deformation of the surrounding rock and ignores the extrusion and pre-convergence deformation of the core surrounding

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rock. However, the surrounding rock deformation or failure often starts with the core rock mass.

Face extrusion is difficult to detect during construction; however, many researchers have confirmed the existence of face extrusion in deep tunnels under excavation unloading through physical model tests or field measurements (Lunardi, 2000; Kamata and Mashimo, 2003; Yoo and Shin, 2003; Wong et al., 2004; Lunardi, 2008). In addition, the rock elements of surrounding rock show principal stress axis rotation before and after the tunnel face passes through, thereby revealing a complex stress path during the tunneling process (Brown and Hoek, 1978; Peila, 1994; Eberhardt, 2001; Jeon et al., 2005). The plastic damage and rock strength weakening are determined primarily by the magnitude of deviatoric stress and direction of the principal stress axis (Carter, 1992; Martin, 1997; Germanovich and Dyskin, 2000). The principal stress rotation may lead to a decrease in rock strength, weakening of physical and mechanical parameters, and change of failure mode (Hoek and Brown, 1980; Eberhardt, 2001; Cai and Kaiser, 2005; Feng et al., 2020), which is especially significant in a bedded or layered rock mass (Zhang et al., 2011a; Li et al., 2020; Li et al., 2021; Li et al., 2022). The deep tunnel excavation, therefore, shows a 3D space effect and significant failures near the tunnel face caused by complex stress path, such as stress magnitude change and stress principal axis rotation (Peila, 1994; Hoek and Marinos, 2000; Lunardi, 2000; Eberhardt, 2001; Jeon et al., 2005; Cantieni et al., 2011; Zhang et al., 2012; Zhang et al., 2019a, 2019b). Existing research has recognized that the complex stress path near the tunnel face is closely related to the surrounding rock stability and construction safety. However, these studies focus on the qualitative evaluation of the changes in the magnitude and direction of principal stress of the surrounding rock elements during the excavation. There is scarce in-depth analysis of the direct causes and mechanical mechanisms of complex stress path or principal stress rotation. Furthermore, there is a lack of quantitative indicators for evaluating the degree of principal stress rotation.

The extrusion deformation of the tunnel face is the most significant manifestation of the 3D space effect, but the in situ measurement using devices, such as a sliding micrometer, is expensive and time-consuming (Lunardi, 2000, 2008). Quantitative research on extrusion deformation mainly involves numerical modeling (Eberhardt, 2001), such as 2D strength evaluation based on the Mohr–Coulomb (Oreste, 2013; Li et al., 2015) and Hoek–Brown (HB) strength criteria (Senent et al., 2013; Li and Yang, 2019). However, these results based on the 2D strength criteria are relatively conservative and differ greatly from the 3D strength criterion (Pan and Hudson, 1988; Kielbassa and Duddeck, 1991; Eberhardt, 2001; Michalowski and Drescher, 2009; Hernandez et al., 2019; Cai et al., 2022a), especially for a deep tunnel where the excavation-induced plastic deformation is widely distributed (Cai et al., 2022a). The excavation process of a deep tunnel often causes a significant 3D stress state around the surrounding rock near the tunnel face (i.e., $\sigma_1 > \sigma_2 > \sigma_3$) (Haimson, 1978; Kang et al., 2010; Chen et al., 2017b; Gong et al., 2019b; Gong et al., 2019a; Feng et al., 2020; Kong et al., 2021). Accurately analyzing the complex stress path and 3D space effect near the deep tunnel face requires reliable 3D numerical models, 3D strength criterion, and constitutive models (Janin et al., 2015; Cai et al., 2022a). Many scholars (Mogi, 1967; Haimson and Chang, 2000; Chang and Haimson, 2005; Mogi, 2006) have observed that the direction of the rock fracture surface is often parallel to the direction of σ_2, which has an important effect on rock strength, deformation, and failure; however, it is often ignored in the selection of a strength criterion for numerical simulation.

This study establishes numerical models of tunnel excavation at different buried depths based on the generalized Zhang–Zhu (GZZ) strength criterion. The change law of 3D extrusion deformation and stress state of tunnels at different buried depths are compared and analyzed. A quantitative index for evaluating the degrees of principal stress rotation is developed to study the stress rotation behaviors of different buried tunnels and the strengthening effects of the core rock mass. Our study can reveal the stress control mechanism and quantitatively evaluate tunnel stability control and construction safety in deep and ultra-deep tunnels, such as the Sichuan–Tibet railway in western China.

2. GZZ strength criterion for deep rock mass

Based on the elastic–plastic analytical method developed by Fama (1993) and Carranza-Torres and Fairhurst (1999) for a circular tunnel, Hoek and Marinos (2000) obtained the statistical relationship between the ratio of rock mass strength to stress (σ_{cm}/σ_v) and the convergence strain of an unsupported tunnel using the Monte Carlo method, as shown in Fig. 1. There is a nonlinear relationship between the convergence deformation of surrounding rock and σ_{cm}/σ_v. The convergence deformation is very sensitive to the rock mass strength at low σ_{cm}/σ_v and approaches zero when σ_{cm}/σ_v is above 0.4.

Theoretically, the stress level or confining pressure of the surrounding rock is in the range of $\sigma_3 = 0$ (tunnel wall) and $\sigma_{3\max} = \min(\sigma_v, \sigma_H, \sigma_h)$ (in situ stress). Hence, $\sigma_{3\max}$ is high for a deep tunnel, and it requires that the applied strength criterion in the numerical simulation accurately reflect the nonlinear behavior of rock strength with stress level to analyze the deformation and stability of the surrounding rock in the deep tunneling. Deep tunnels are often in a high-ground stress environment (high vertical stress σ_v), and therefore result in a small σ_{cm}/σ_v and large convergent deformation (Fig. 1). The refined analysis of deep surrounding rock stability requires a rock strength criterion of high applicability and reliability because the convergent deformation of surrounding rock is very sensitive to the rock mass strength (σ_{cm}) at low σ_{cm}/σ_v in a deep tunnel.

Zhang and Zhu (Zhang and Zhu, 2007; Zhang, 2008) proposed a generalized 3D nonlinear strength criterion for engineering rock mass (GZZ strength criterion) by combining the advantages of the 3D Mogi strength criterion (Mogi, 1967, 2006) and 2D HB strength criterion (Hoek et al., 2002), which is given in Eq. (1). This GZZ strength criterion can accurately describe the nonlinear transformation of the triaxial tension state to the triaxial compression state in the 3D stress space on the π-plane.

$$\frac{1}{\sigma_c^{(1/a-1)}}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)^{1/a} + \frac{m}{2}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right) - \frac{m}{2}(\sigma_1 + \sigma_3) - s\sigma_c = 0 \tag{1}$$

where σ_c is unconfined compressive strength; m, s, and a are strength parameters for rock masses; τ_{oct} is octahedral shear stress.

The relationship between principal stress and stress invariants is as follows:

$$\begin{bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\end{bmatrix} = \frac{2}{\sqrt{3}}\sqrt{J_2}\begin{bmatrix}\sin\left(\theta_\sigma + \frac{2}{3}\pi\right)\\ \sin(\theta_\sigma)\\ \sin\left(\theta_\sigma - \frac{2}{3}\pi\right)\end{bmatrix} + \begin{bmatrix}I_1/3\\ I_1/3\\ I_1/3\end{bmatrix} \tag{2}$$

where J_2 is second and third deviatoric stress invariant; θ_σ is lode angle.

Substituting Eq. (2) into Eq. (1), we obtain the GZZ strength criterion in Eq. (3), which is expressed using the stress invariant.

$$\frac{1}{\sigma_c^{(1/a-1)}}\left(\sqrt{3J_2}\right)^{1/a} + \left(\frac{3+2\sin\theta_\sigma}{2\sqrt{3}}\right)m_b\sqrt{J_2} - m_b\frac{I_1}{3} - s\sigma_c = 0 \tag{3}$$

Cai et al. (2021) further modified the GZZ strength criterion to overcome the non-smoothness geometric characteristics of the original GZZ with the following mathematical expression:

$$F = \frac{1}{\sigma_c^{(1/a-1)}}\left(\sqrt{3J_2}\right)^{1/a} + \left(\frac{3+\sin3\theta_\sigma}{2\sqrt{3}}\right)m_b\sqrt{J_2} - m_b\frac{I_1}{3} - s\sigma_c = 0 \tag{4}$$

The modified and original GZZs in Eqs. (4) and (3), respectively,

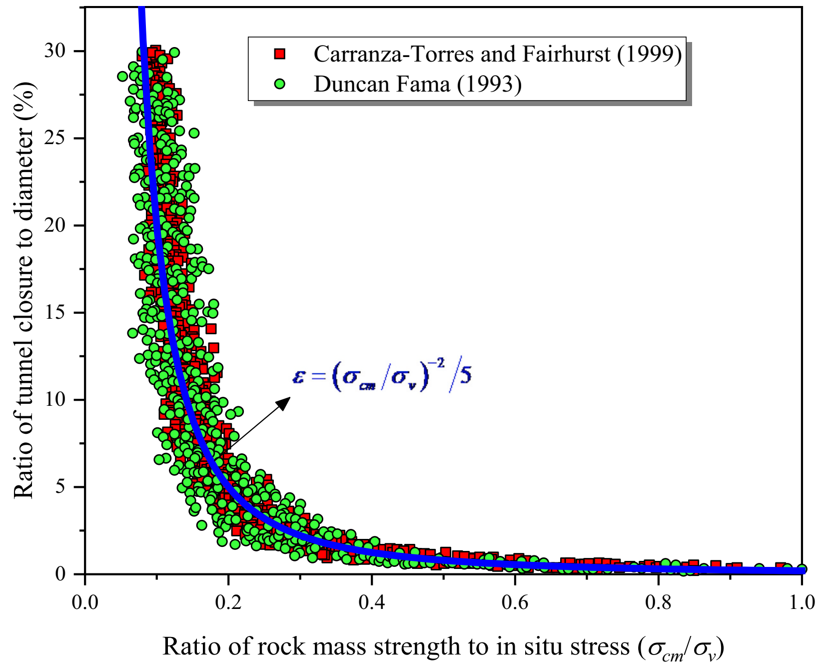


Fig. 1. Relationship between the convergence deformation of surrounding rock with σ_{cm}/σ_v (ratio of strength to stress) (Hoek and Marinos, 2000).

have similar mathematical and geometric characteristics on the π-plane (Fig. 2). There are three intersections in each criterion, at $\theta_\sigma = \pm \pi/6$ and 0, providing smoothness and derivability of numerical calculations. Meanwhile, the new GZZ has better adaptability for use in a rock mass with low quality (i.e., soft rock) and high stress conditions (i.e., deep rock engineering) (Cai et al., 2021). Therefore, it can be used in deep soft rock engineering under high ground stress.

The main parameters in HB and GZZ are m_b, s, and a, which reflect the structural and mechanical properties of in situ rock mass, as follows:

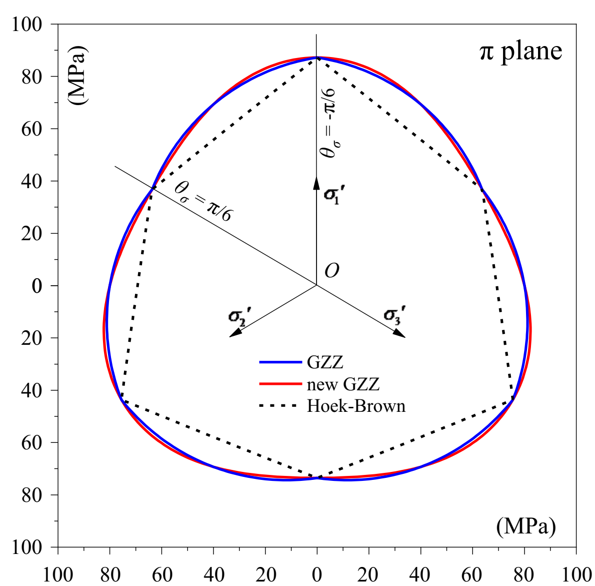


Fig. 2. Strength envelopes of different criteria on the π-plane [strength parameters come from (Cai et al., 2021)].

$$\begin{cases} m_b = m_i exp\left(\dfrac{GSI - 100}{28 - 14D}\right) \\ s = exp\left(\dfrac{GSI - 100}{9 - 3D}\right) \\ a = \dfrac{1}{2} + \dfrac{1}{6}[exp(-GSI/15) - exp(-20/3)] \end{cases} \tag{5}$$

where m_i is the indoor rock parameter (Hoek et al., 2002); GSI is the geological strength index and reflects the structural information (surface condition and structure), which can also be transferred from other rock classification system, e.g., rock mass rating (RMR) (Zhang et al., 2019a, 2019b); D is the excavation disturbance coefficient, related to the rock mass integrity coefficient.

The new GZZ strength criterion has high fitting accuracy for true triaxial test data (Cai et al., 2021) and can directly adapt the parameters obtained by the HB criterion (Cai et al., 2021; Cai et al., 2022a). This is the foundation of this study to conduct dynamic and forward numerical analyses.

3. 3D numerical forward analysis for rock tunnel excavations of different depths

This section focuses on the 3D space effect near the tunnel face during excavation. The 3D finite element numerical method is used to model the tunnel excavation process using the new GZZ strength criterion. The rock mass rating (RMR) and GSI, which evaluate rock mass structure, can be dynamically reflected by the advancing tunnel face information (Chen et al., 2017a; Chen et al., 2021a; Chen et al., 2021b; Cai et al., 2022a); therefore, the digital in situ detection of tunnel face information is applied to obtain the rock strength parameters. Note that the commonly applied parameter acquisition methods (e.g., laboratory rock test and field deformation-based back analysis) could not meet the requirements of forward and dynamic analyses of tunneling construction. This study intends to provide a link between geological rock mass conditions and engineering tunneling simulations.

3.1. Engineering geology

The rock strata and rock mass parameters are selected from the geological exploration data of a typical deep tunnel in Western China using the on-site digital intelligent acquisition method. The DXG is the "deepest" highway tunnel with a maximum buried depth of 1944 m and length of 12.1 km. The surrounding hard rock is moderately weathered dolomite with a saturated uniaxial compressive strength (UCS) of 140 MPa. The integrity coefficient of the tunnel face obtained by numerical acquisition method is $K_v = 0.65$. The lateral pressure coefficients in the maximum and minimum horizontal geostress directions are 1.32 and 0.874, respectively, and the angle between the maximum horizontal geostress and tunnel axis is 52° (Fig. 3a). The tectonic stress of the tunnel site is complicated. The main structure comprises a series of north-south folds and faults, which are accompanied by east-west extensional or extensional-torsional faults and northwest or northeast torsional faults. The major fault that crosses the DXG tunnel is a reverse fault >19 km in length that extends in the northeast direction and tends to the southeast direction at an incline of 70°–80°. The fracture zone of the fault is granular and severely compressed. High ground stress-induced disasters, such as distortion of steel arch, rock burst, and large deformation, occur frequently near the deep tunnel face and rarely far away from the tunnel excavation face. Therefore, the 3D space effect near the deep tunnel face impacts the surrounding rock stability and construction safety.

3.2. 3D numerical modeling of tunnel excavation

A 3D geological model is established according to the geological survey report and field geological information, as shown in Fig. 3b. Because the long length makes it difficult to model the whole tunnel excavation process, a representative tunnel section of size x300 × z300 × y380 m is selected for 3D numerical analysis (Fig. 3c). The tunnel excavation face is in the *x*-*z* plane, and its advancing direction (tunnel axis) is parallel to the *y*-axis. The numerical meshing model adopts a tetrahedral element, and the three-level division is adopted to balance the calculation time and accuracy. The mesh size is 0.6 m for the tunnel, 4 m for the circular transition zone outside the tunnel, and 20 m for the model boundary. The radiation ratio of adjacent mesh division used to control the mesh generation for the planar area with an inner boundary is set as 0.3. The tunnel geometry and support parameters follow the engineering design data.

The comprehensively obtained initial stress from the field measurement and geological survey report is transformed into the load boundary conditions [Fig. 3(c.2)] of the numerical model to match the tunneling direction. When the in-situ stress equilibrium condition is satisfied, the model boundary conditions are changed to a fixed constraint [Fig. 3 (c.2)] to retain the initial in situ stress field. The tunnel excavation is simulated after setting the initial geostress field. The excavation length is 100 m, and cyclic footage is 2.0 m. Each cyclic footage includes two construction steps (excavation and support steps); Fig. 3(c.3) shows the tunnel lining structure based on the design data. The excavation processes of the tunnels with depths ranging between 200 and 2000 m are simulated with depth increments of 200 m.

3.3. Numerical intelligent acquisition of rock mass parameters

A 3D nonlinear GZZ strength criterion having similar input parameters as the HB criterion, including indoor rock mechanics parameters

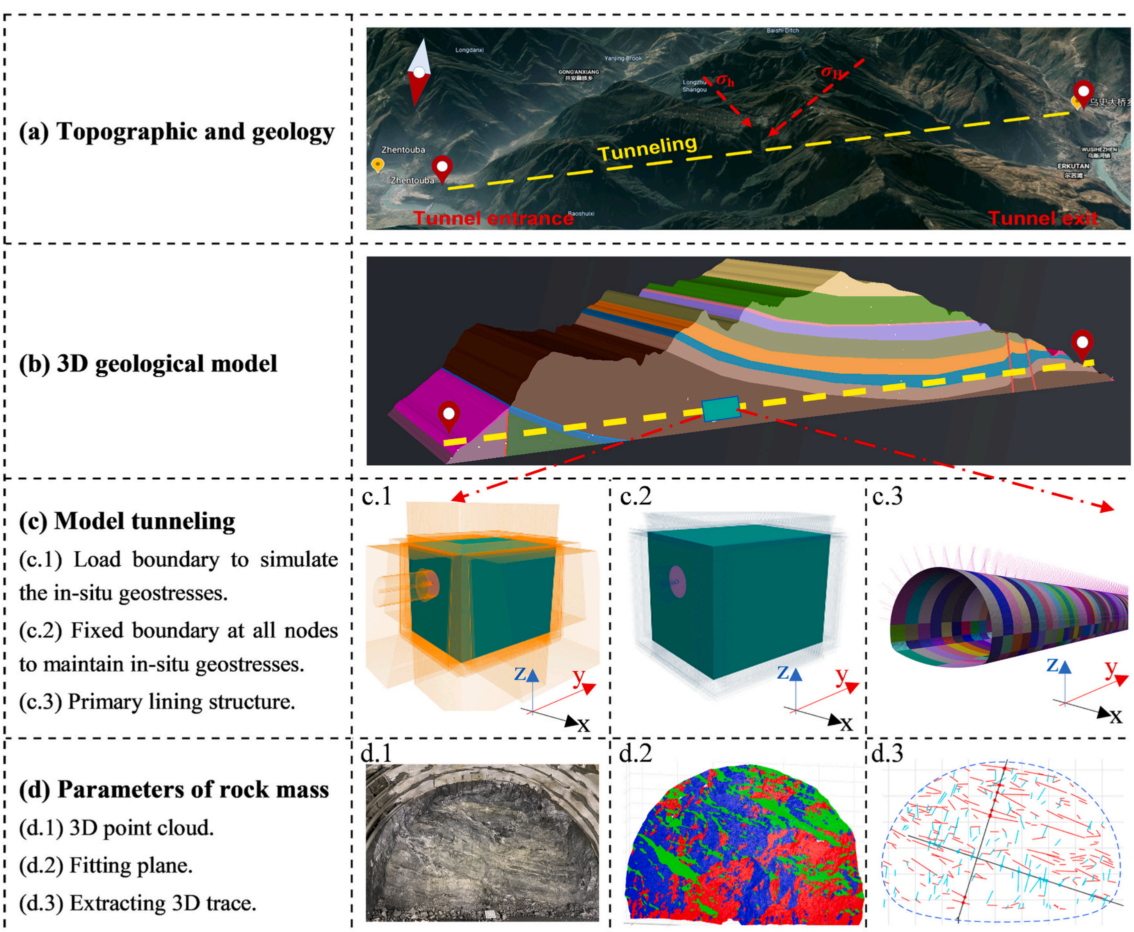


Fig. 3. Refined 3D numerical simulation for rock tunnel excavation.

and in situ rock structure parameters, is used for numerical analysis. The key parameters are the indoor rock UCS (σ_{ci}), and on-site rock mass discontinuity parameters *GSI*, and *D*. The accuracy of these parameters will directly impact the numerical analysis.

3D laser scanning and reconstruction methods (Chen et al., 2017a; Zhu et al., 2017; Li et al., 2019; Zhang et al., 2020) are applied to extract structural plane geometric information (e.g., joints and fissures), as shown in Fig. 3d. 3D point cloud information (Fig. 3d) is extracted using mobile phone photography or 3D laser scanning. Next, a 3D reconstruction method acquires the geometric information of 3D trace occurrence, group number, and distribution characteristics of the tunnel face [Fig. 3(d.2–3)]. Finally, rock mass parameters, such as *GSI* and *D,* are obtained according to approaches recommended by Hoek et al. (Hoek and Marinos, 2000; Hoek et al., 2002; Hoek and Brown, 2019) [i.e., Eq. (5)]. Thus, the input parameters for 3D numerical calculation are obtained by integrating the geological exploration report, indoor rock test results, and intelligent acquisition of in situ rock structure information, as shown in Table 1.

Table 1 lists the parameters of a typical medium quality rock mass (GSI = 46). The parameter *D* is related to structural joints, non-structural joints, primary structure, and secondary structure. Its accurate value is difficult to obtain in engineering and geological fields, and is set as $D = 1 - K_v = 0.35$ in this study.

4. Extrusion deformations and stress conditions of tunnel face under different buried depths

This study focuses on the extrusion deformation of the tunnel face. The most effective method to measure the extrusion deformation is the sliding micrometer (Lunardi, 2000, 2008). However, the construction operation of extrusion monitoring by sliding micrometer is difficult due to construction safety of the tunnel face (e.g., severe rock burst of DXG deep tunnel). The reliability and accuracy of the suggested digital methods given in Fig. 3 refer to the comparisons between the numerical results and monitoring displacements conducted by Cai et al. (2022a).

4.1. Longitudinal variation law of surrounding rock stress state

Fig. 4 shows the stress state distribution of the 2000 m deep DXG tunnel. The stress state of rock or rock mass during excavation is evaluated using the yield approach index (*YAI*) (Zhang et al., 2011b). The *YAI* indicator for GZZ is developed as (Cai et al., 2022a,b):

$$YAI_{\mathrm{GZZ}} = \begin{cases} \dfrac{J_2(\sigma_1,\sigma_2,\sigma_3)}{J_2(I_1,\theta_\sigma)\,|_{\mathrm{GZZ}}}, & J_2(\sigma_1,\sigma_2,\sigma_3)\langle J_2(I_1,\theta_\sigma)\,|_{\mathrm{GZZ}} \\ 1+\dfrac{\gamma^p}{\gamma^p_{\max}}, & J_2(\sigma_1,\sigma_2,\sigma_3)\geq J_2(I_1,\theta_\sigma)\,|_{\mathrm{GZZ}} \end{cases} \tag{6}$$

where $J_2(\sigma_1, \sigma_2, \sigma_3)$ is the second invariant of deviatoric stress; $J_2(I_1, \theta_\sigma)|_{\mathrm{GZZ}}$ is the radius of GZZ strength envelope on the π-plane; and YAI_{GZZ} is used to evaluate the yield or failure state of rock elements quantitatively. The rock is in elastic stress state with $YAI_{\mathrm{GZZ}} < 1$ and $YAI_{\mathrm{GZZ}} = 1 + \gamma^p/\gamma^p_{\max} \geq 1$ for the plastic stress state, where γ^p is the plastic shear strain and is given as:

$$\gamma^p = \sqrt{\left(\varepsilon^p_{ij} - \varepsilon^p_m\delta_{ij}\right)\left(\varepsilon^p_{ij} - \varepsilon^p_m\delta_{ij}\right)/2} \tag{7}$$

The rock stress state located near the coordinate origin on the π-plane (point *C* in Fig. 4) shows a hydrostatic stress state (obtained and transformed from field measurement and geological survey report) at the tunnel section far away from the tunnel face, where *YAI* is close to 0, and in an elastic stress state. At the excavated and stabilized tunnel section (point *A* in Fig. 4), the radius of the GZZ envelope on the π-plane is smaller than that of the unexcavated rock (point *C* in Fig. 4). The surrounding rock is in the stress unloading state, i.e., the plastic yield or failure state with its stress point located outside the GZZ strength envelope. At the tunnel face section (point *B* in Fig. 4), the stress state of the rock crown element is near the GZZ envelope line with *YAI* close to 1.0, thereby indicating a critical elastic-plastic stress state. The radius of the strength envelope at point *B* is larger than that at points *A* and *C*, thereby indicating stress loading conditions in the rock around the tunnel face.

The *YAI* longitudinal distribution curves in Fig. 4 show that the surrounding rock gradually changes behavior from the elastic stress state (*C*) of unexcavated rock to the plastic or failure stress state (*A*) of excavated rock; however, the change is drastic near the tunnel face because of the significant 3D space effect, and the stress state and deformation characteristics are significantly different from the plane strain tunnel model. Because the *YAI* of the tunnel floor at the excavated section is the largest and shows a plastic yield or failure state, the floor heave deformation of the excavated tunnel should be controlled and the tunnel invert should be closed in time to ensure sufficient operational stiffness and strength.

4.2. Stress and strength behaviors of tunnel face rock

Fig. 5 shows the stress states of rock units and a distribution area of *YAI* < 2.0 at the tunnel face section, where the boundary of the red area (dotted line) is the isoline of *YAI* = 2.0, and the unstable area is similar to the tunnel contour.

The face center first appears as the yield or failure state at the section of the tunnel face. The rock at the face center is the most unstable point, with a *YAI* much larger than that of the surrounding rock. The boundary of the tunnel face (green area in Fig. 5) shows a relatively stable state with *YAI* < 2.0. Based on these results, we confirm that instability first appears at the tunnel face.

The strength envelope of core rock is much smaller than that of the surrounding rock, showing noticeable stress unloading and unstable tensile stress in the direction of σ_3. The instability states, as obtained from Fig. 5, decrease along the tunnel face center (*YAI* = 2.07), tunnel floor (*YAI* = 1.07), tunnel crown (*YAI* = 0.98), and tunnel sidewall (*YAI* = 0.77). Therefore, it can be considered that the tunnel face stability is worse than that of the surrounding rock, and the tunnel floor stability is the worst among the surrounding locations, which corresponds to the *YAI* longitudinal distribution in Fig. 4.

4.3. Stress and strength behaviors of the tunnel face rock at different buried depths

The radius of the GZZ strength envelope on the π-plane gradually increases with increasing buried depth (Fig. 6a). The stress state gradually transfers from the inner to the outer envelope. Deeper buried depth corresponds to farther distance between the stress state with high strength envelope and core rock mass with the worst stability. σ_3 of rock elements are almost zero because of the supported tunnel face. As σ_3 approaches zero, the unloading effect becomes more pronounced with increasing buried depth, and σ_3 changes from compression to tensile stress at buried depths of >1600 m.

The stability of surrounding rock can be comprehensively estimated by the *YAI*, plastic proportion, and rock deformation. These three

Table 1
Physical and strength parameters of surrounding rock mass.

Lithology	ρ (kg/m^3)	*E* (MPa)	*N*	σ_c (MPa)	*GSI*	*D*	K_v	m_i	*c* (MPa)	φ (°)
Dolomites	2650	6270	0.28	140	46	0.35	0.65	11	3.93	31.7

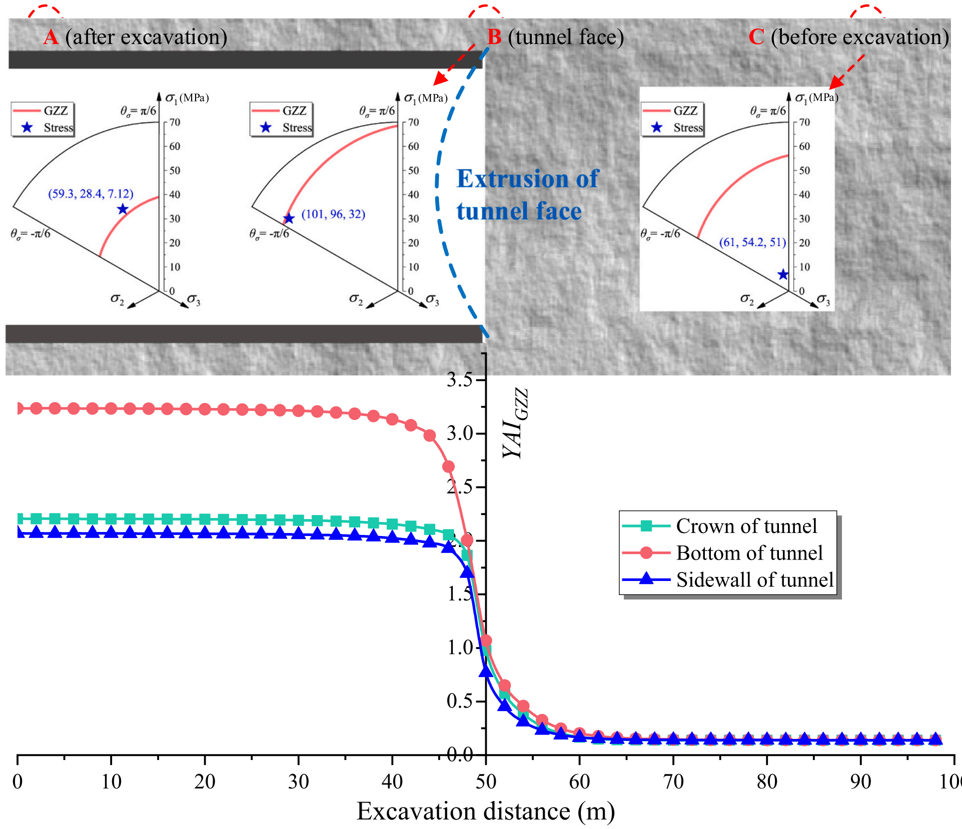


Fig. 4. Stress states of surrounding rock and longitudinal YAI_{GZZ} distributions.

physical quantities are extracted to evaluate the stability of the tunnel face, as the rock at tunnel face often exhibits the worst stable state compared with the surrounding rock (tunnel crown, sidewall, and floor). The rock elements in the tunnel face center, with buried depths of <400 m, are in elastic stress state (*FAI* < 1.0 in Fig. 6(a)) due to the high UCS of the rock. This leads to the inference that the surrounding rock is also in the elastic stress state because the stability of the tunnel face is worse than that of the surrounding rock. For tunnel depths deeper than 400 m, the *FAI* is larger than 1.0 and gradually increases as buried depth increases. When the buried depth reaches 2000 m (ultra-deep tunnel), the tunnel face may be damaged or collapse with *FAI* > 2.0.

The proportion of plastic deformation increases significantly in a shallow tunnel, and it gradually reaches 80% for a tunnel with a buried depth >800 m, thereby indicating that the deep tunnel is almost affected by plastic deformation. The extrusion deformation of the tunnel face increases significantly with increasing buried depth because of the nonlinear plastic deformation, which confirms the significant 3D space effect of deep tunnel operation.

5. Complex stress path and principal stress rotation in deep tunnel excavation

5.1. Principal stress rotation of surrounding rock in deep tunnel excavation

The stress state and deformation characteristics near the tunnel face are significantly different from those of more stable sections far away from the tunnel face (often regarded as plane strain tunnel). In this study, the complex stress path and principal stress rotation behaviors are examined as the tunnel face approaches and passes through the rock elements at the crown and sidewall. It should be noted that the influence of the tunnel support structure has been considered in numerical simulation, and the size of rock element was set as 0.6 m near the tunnel wall. Therefore, the minimum principal stress (σ_3) would not be reduced to zero during excavation.

As shown in Fig. 7a-b, as the excavation face gradually approaches and passes through the rock units (rock unit *A* for tunnel crown and rock unit *B* for tunnel sidewall), σ_1 and σ_2 of rock units gradually increase (i.e., stress loading) and σ_3 decreases (i.e., stress unloading). The gradually increasing differential stress contributes to the surrounding rock instability. The principal stress changes significantly within 2 m of the tunnel face and exhibits the principal stress rotation phenomenon. For the rock element at the tunnel crown (Fig. 7a), the direction of σ_1 is originally in line with σ_x, but rotates by 90° in the direction of σ_1 and σ_2 within 2 m before and after the excavation (± 2 m). σ_1 transfers from the *x*- axis to *y*-*z* plane at a certain angle with the *y*- and *z*- axes; the direction of σ_2 is initially in the *y*-*z* plane but abruptly transfers in the direction of the *x*-axis (i.e., σ_x). For the rock element at the sidewall (Fig. 7b), σ_1 transfers from σ_x before excavation to σ_z after excavation. In contrast, σ_2 transfers from σ_z before excavation to σ_x after excavation; σ_3 produces 90° rotations after the excavation.

The red dotted line in Fig. 7a-b denotes the triaxial compressive strength (TCS) calculated using the GZZ strength criterion by fixing σ_2 and σ_3. The rock is assumed to be in the plastic yield state if σ_1 is higher than the TCS. The surrounding rock stability condition during tunneling can be judged by comparing the relative difference between the TCS and σ_1, which contributes to conducting engineering support design and stability control. To ensure the surrounding rock stability, passive supports, such as an anchor and steel arch frame, can be used to drive σ_1 below the TCS. The blue dotted line in Fig. 7a-b is the rock failure confining pressure (FCP), which is determined using the GZZ strength criterion by fixing σ_1 and σ_2. The rock is assumed to be in the plastic state if unloading stress σ_3 is lower than FCP (i.e., σ_3 < FCP). For ensuring tunnel stability, active supports, such as suppressing the extrusion deformation of the tunnel face (i.e., New Austrian method) or strengthening the core rock mass (i.e., ADECO-RS method), are used to

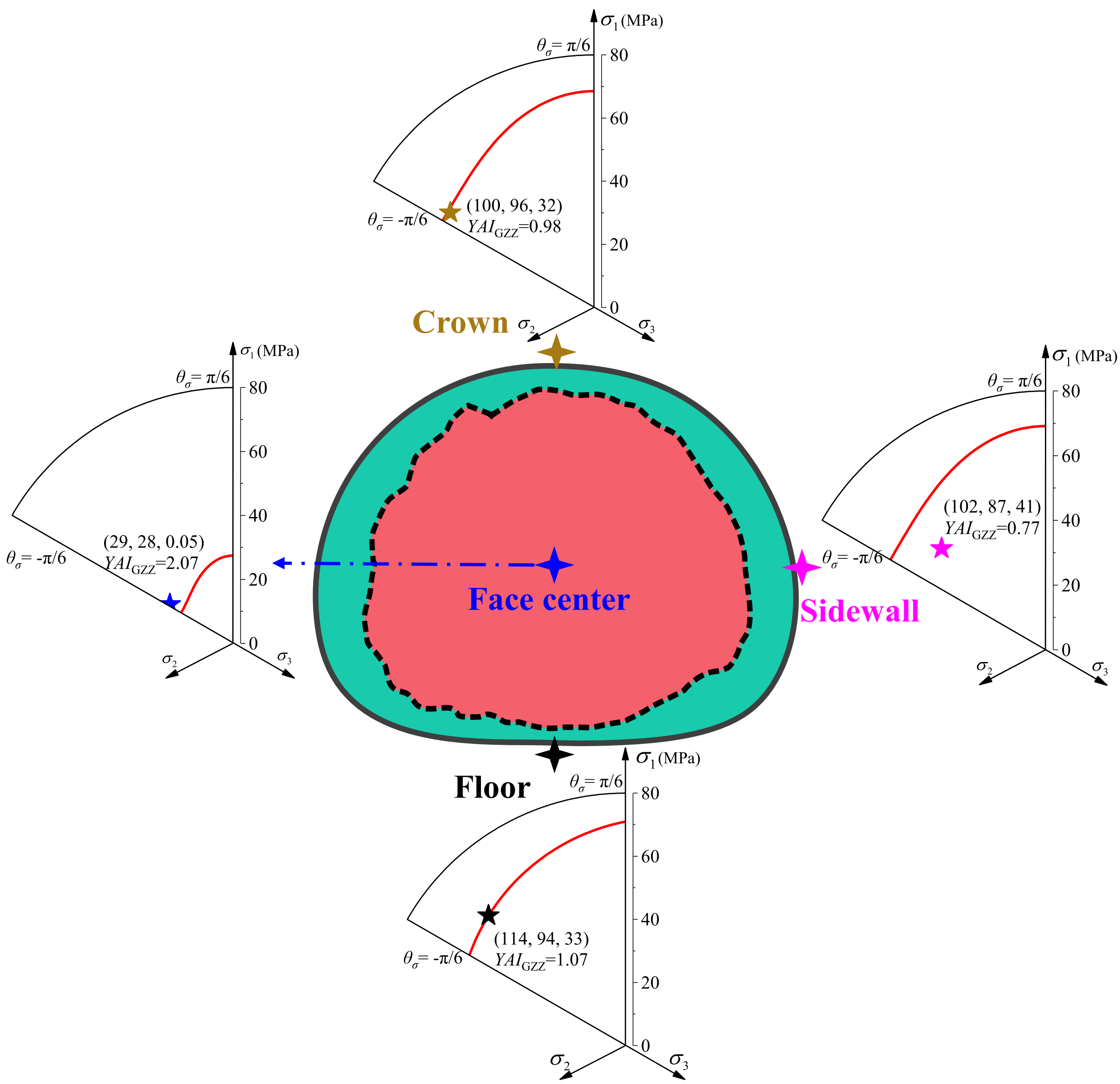


Fig. 5. Stress states and GZZ strength envelopes on π-plane of surrounding and tunnel face center rock elements.

control the deformation of core surrounding rock and restrain the stress unloading caused by excavation and realize the stability by putting σ_3 values at or above the FCP line.

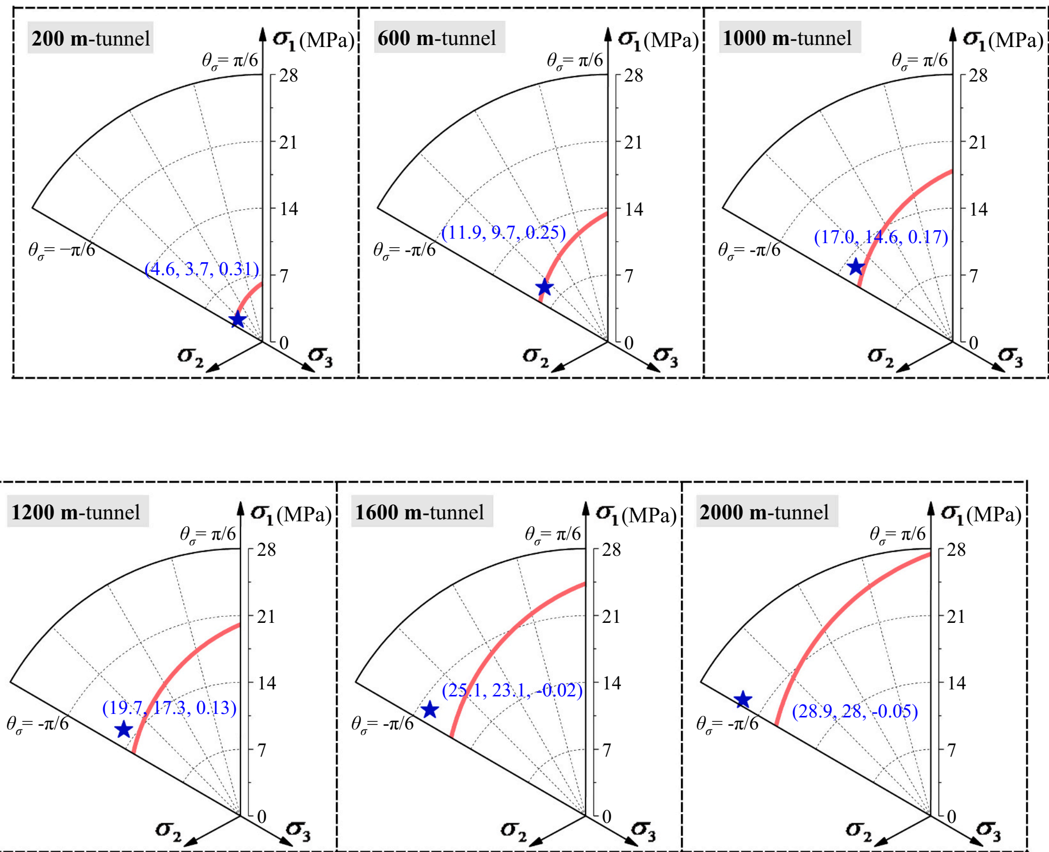
Before the tunnel face reaches the rock unit in the elastic stress state, the principal stress of the rock is also rotated. This, however, does not weaken the rock mass parameters because of the reversibility of elastic deformation and stress. However, the rock is in the plastic stress state when the tunnel face passes through, and the principal stress rotation weakens the rock strength parameters; hence, it is necessary to focus on the stability of the surrounding rock within 2 m behind the excavation face. The potential strike and tendency of the rock failure surface are found to be parallel to the directions of σ_2 and σ_3, respectively (Mogi, 1967; Haimson and Chang, 2000; Chang and Haimson, 2005; Mogi, 2006; Zhang et al., 2011a); therefore, the information in Fig. 7a-b can be used as reference when designing the support direction.

As shown in Fig. 7a-b, the difference between σ_1 and TCS is relatively large when the rock is in the plastic state. Hence, the implementation of a passive control method requires that σ_1 be reduced by >10 MPa. However, because the difference between σ_3 and FCP is small, the implementation of the active control method requires that σ_3 be increased by <2 MPa. Therefore, the implementation of the active control method is reasonable and effective. Furthermore, σ_1 is highly sensitive to changes in σ_3 at small values of σ_3. Therefore, a slight increase in σ_3 could greatly enhance the rock strength; however, the strengthening effect is limited when σ_3 is large. Hence, it is efficient to use the active control method for rock masses with small confining pressure.

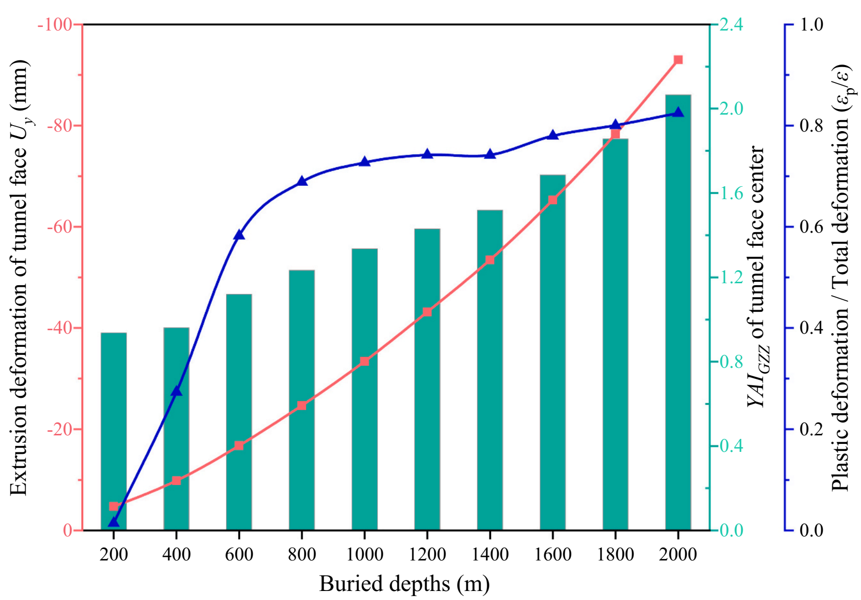
5.2. Influence mechanism of principal stress rotation on rock stability

5.2.1. Decrease strength and weaken strength parameters of rock mass

The complex stress path caused by tunnel excavation is the controlling factor for deep tunnel stability. The magnitude and direction of the principal stress near the tunnel face decrease the rock strength, weaken the rock parameters, and change the failure mode (Hoek and Brown, 1980; Eberhardt, 2001; Cai and Kaiser, 2005; Feng et al., 2020). There are no secondary cracks in the rock element under the original ground stress, but when the redistribution of the stress magnitude and direction caused by excavation unloading exceeds the damage threshold or the yield strength of the rock (rock mass), microcracks may occur (Fig. 8). Moreover, each excavation step causes principal stress rotation and generates new microcracks, gradually increasing the crack density and depth, and eventually connecting the new and old cracks. The changes in stress magnitude and direction contribute to the depth and



(a)

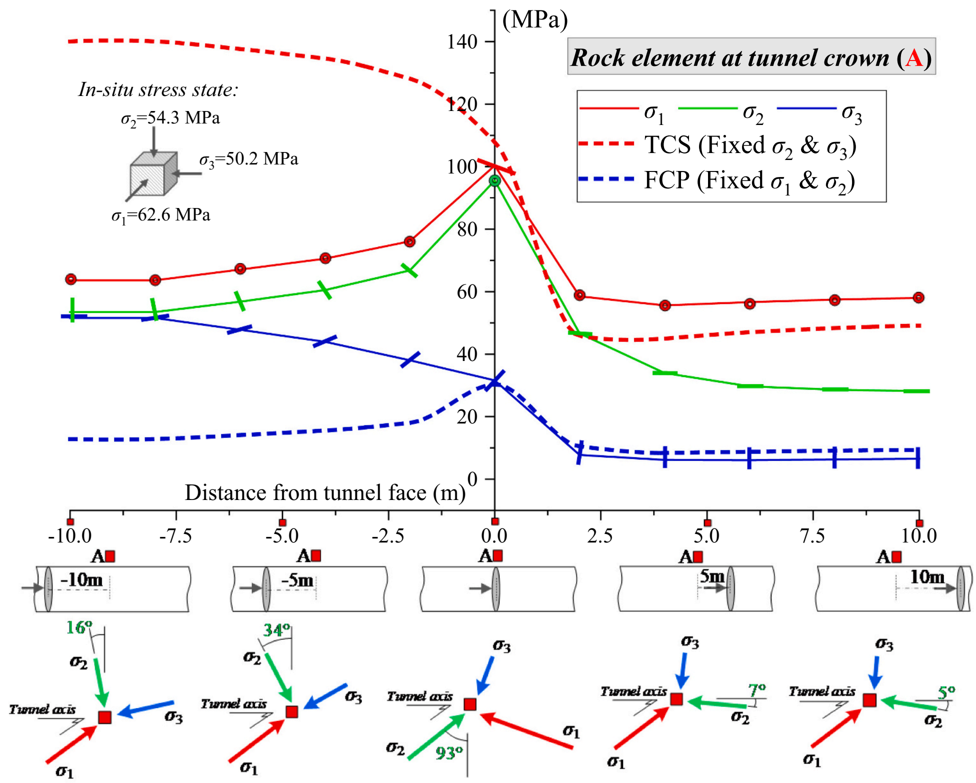


(b)

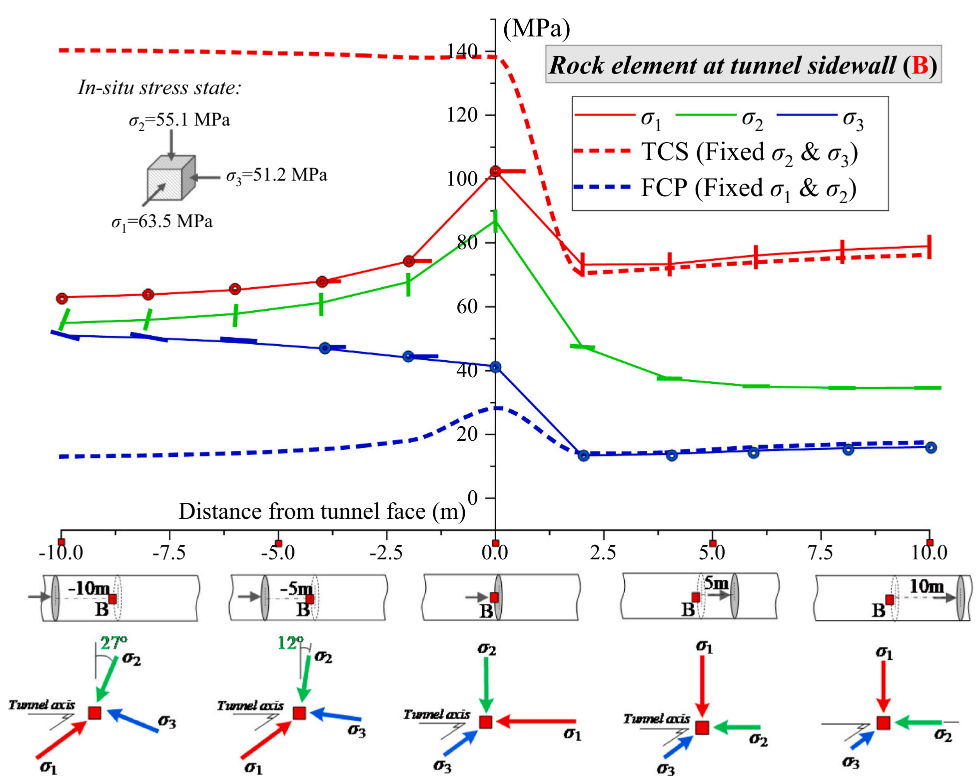
Fig. 6. Stress and strength behaviors of tunnel face elements at different buried depths. (a) core rock stress state on the π-plane; (b) core rock failure and deformation evaluation.

density of cracks, respectively, and the combined factors contribute to the rock failure. Once the initial cracks appear, the stress rotation impacts the direction and depth of crack expansion, thus gradually weakening the rock parameters.

As shown in Fig. 9, I_1 and J_2 reflect the stress levels and deviations of the rock mass on the π-plane, respectively, and θ_σ reflects the stress state. The change in stress magnitude can reduce the rock strength and change the stress state from the inner to the outer π-plane during excavation. However, changes in the stress direction will weaken the rock strength parameters (i.e., m_b, s, a) (Fig. 8), thereby gradually shrinking the



(a) Principal stress rotations of rock unit at tunnel crown



(b) Principal stress rotations of rock unit at tunnel sidewall

Fig. 7. Principal stress rotations of the surrounding rock units caused by excavation in 2000 m buried deep tunnel. (a) Principal stress rotations of rock unit at tunnel crown. (b) Principal stress rotations of rock unit at tunnel sidewall.

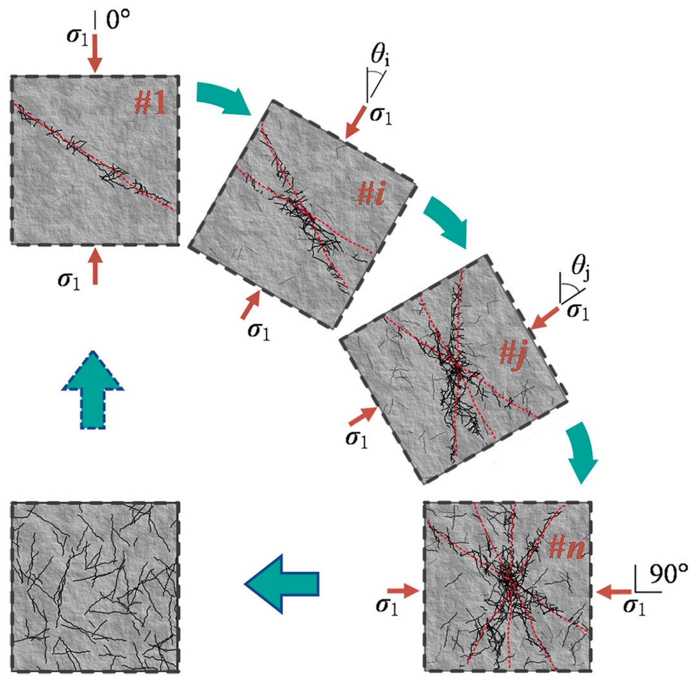


Fig. 8. Schematic of rock strength parameter weakening caused by principal stress rotation.

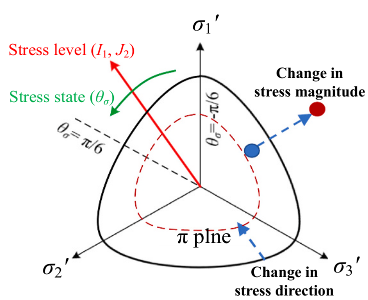


Fig. 9. Illustration of the principal stress rotation contributing to the decrease in rock strength and weakening of strength parameters.

strength envelope (by decreasing the radius) from the black envelope to the red dotted envelope.

5.2.2. Change in failure mode of rock mass

Experimental results show that σ_2 and σ_3 control the strike and tendency of fracture surfaces, respectively (Mogi, 1967; Haimson and Chang, 2000; Chang and Haimson, 2005; Mogi, 2006; Zhang et al., 2011a). The excavation process is often accompanied by significant stress rotation, as shown in Fig. 7a-b. The changes in the directions of σ_2 and σ_3 subsequently result in the change of failure mode (angle of failure plane, tensile failure, or shear failure). In addition, the rock fracture surface may also produce corresponding rotation if the rock reaches its failure strength. Hence, the principal stress rotation may change the failure mode of the rock mass and produce a plastic deformation increment in the plastic calculation theories (Zheng, 2002).

5.2.3. Influence of stress rotation on tunnel stability excavated in bedded or large-size geometric discontinued rock mass

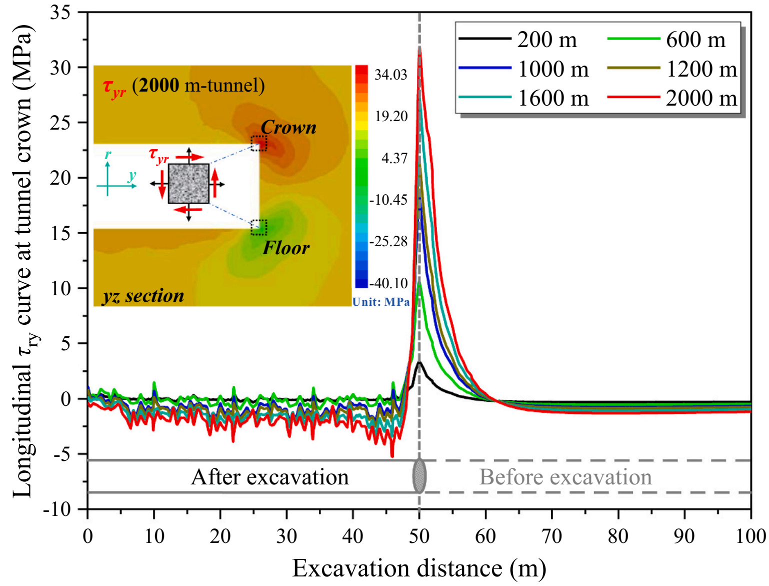
The degree of stress rotation is relatively significant near the deep tunnel face. For the deep tunnel excavated in bedded rock or passing through the fault, it is easy to activate the fault when the σ_3 rotates to the direction perpendicular to the fault plane and σ_1 exceeds the shear strength of the fault plane, subsequently resulting in fault slipping along the plane. Therefore, for deep buried tunnels with faults or obvious joint surfaces, there is higher possibility of instability or failure caused by stress rotation as compared with homogeneous and isotropic rock strata (Zhang et al., 2011a; Li et al., 2020; Li et al., 2022). This may result in severe consequences if large-size geometric discontinuities are not considered as the stress rotates during tunneling.

5.2.4. Difference of principal stress rotation on deep and shallow tunnels

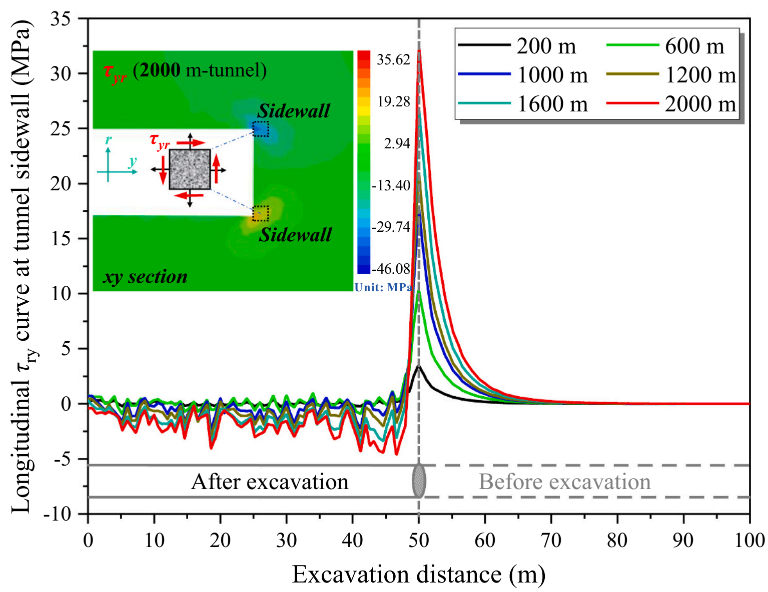
Most researchers have ignored the role of the complex stress path and principal stress rotation in the design and construction of shallow tunnels because the low-stress magnitude is below the damage threshold or yield strength, and the rotation does not induce new cracks. Hence, principal stress rotation in a shallow tunnel may only decrease the rock strength without weakening rock parameters (strength envelope shrinkage in Fig. 9) and changing the failure mode. However, for a deep rock tunnel with high geostress, the excavation unloading produces a large plastic zone, and the change in stress magnitude decreases the rock strength. The change in stress direction will continuously increase the fracture density, weakening the rock parameters and changing the failure mode (changes in σ_2 and σ_3). Hence, the influence of principal stress rotation on the deep tunnel is much more severe than that in the shallow tunnel, thus restricting the rock stability and construction safety in the deep tunnel operation.

5.3. The reason for principal stress rotation and shear stress analysis

The principal stress near the tunnel face is inconsistent with the plane strain coordinate system (r, θ, y) owing to the principal stress rotation, where r, θ, and y are the radial, tangential, and axis (advancing) directions of the tunnel, respectively. The shear stress in the deep tunnel increases with increasing principal stress (Zheng, 2002). The shear stress τ_{ry} (τ_{yz} for tunnel crown and τ_{xy} for sidewall) is almost zero (Fig. 10) at the tunnel section outside the 5 m range from the tunnel face, but shows a sharp change near the tunnel, increasing from 0 to 30–35 MPa. The superposition of τ_{ry} and normal stress in the plane strain coordinate system rotates the principal axis, and the principal stress near the tunnel face shows a certain deviation from the radial and axial directions of the tunnel.



(a) τ_{ry} (τ_{zy}) shear stress for tunnel crown



(b) τ_{ry} (τ_{xy}) shear stress for tunnel sidewall

Fig. 10. τ_{ry} of surrounding rock units: (a) tunnel crown and (b) tunnel sidewall.

As shown in Fig. 10, the fluctuation of τ_{ry} in the deep tunnel is more intense. τ_{ry} is almost zero in shallow tunnels (e.g., 200 m buried tunnel), and gradually increases with increasing buried depth. Traditional shallow tunnel design and construction mainly concern the normal stress in the (r, θ, y) direction and ignore the importance of shear stress on the tunnel stability. τ_{ry} directly contributes to the complex stress path and principal stress rotation near the tunnel face; therefore, it should be considered seriously in design and construction of the deep tunnel.

5.4. Quantitative index for evaluating the principal stress rotation

Quantitative evaluation of the weakening law of the complex stress path (stress rotation) on rock mass parameters remains challenging because of the difficulties in experiment verification. There is scarce research in qualitative investigation of the principal stress rotation near the tunnel face; a convenient and applicable quantitative evaluation index is still needed. In this section, the angle index, shear stress increment index, and shear stress index will be developed and compared to quantitatively evaluate the degree of principal stress rotation.

5.4.1. Angle index

Rotation angle is the most direct evaluation index. The direction of the σ_i ($I = 1, 2, 3$) can be expressed as $\mathbf{r}^k = (l_i^k, m_i^k, n_i^k)$, where ($l$, m, n) represents the cosine of the angle between the principal stress and (x, y, z).

The rotation angle of σ_i between the k and $k+1$ excavation footage is shown in Eq. (8).

$$d\theta_i = 2arcsin\left[\frac{\left|\mathbf{r}_i^{k+1}-\mathbf{r}_i^{k}\right|}{2}\right] = 2arcsin\left[\frac{\left|\overrightarrow{\left(l_i^{k+1},m_i^{k+1},n_i^{k+1}\right)}-\overrightarrow{\left(l_i^{k},m_i^{k},n_i^{k}\right)}\right|}{2}\right] \tag{8}$$

The angle index $g(\theta)$ is obtained by normalizing the rotation angles in the three principal stresses as follows:

$$g(\theta) = \frac{\sqrt{(d\theta_1)^2+(d\theta_2)^2+(d\theta_3)^2}}{\sqrt{3}\pi} \tag{9}$$

5.4.2. Shear stress increment index

The sharply increasing τ_{ry} is the direct cause of principal stress rotation near the tunnel face (Section 5.3); hence, the shear stress increment between adjacent excavation steps can be used to evaluate the degree of principal stress rotation.

The shear stress vector and shear stress increment between two adjacent excavation steps are defined as:

$$\begin{cases} \boldsymbol{\tau} = \boldsymbol{\tau}(y) = \overrightarrow{\left(\tau_{xy},\tau_{yz},\tau_{xz}\right)} \\ d\boldsymbol{\tau} = \overrightarrow{\left(\tau_{xy}^{k+1},\tau_{yz}^{k+1},\tau_{xz}^{k+1}\right)} - \overrightarrow{\left(\tau_{xy}^{k},\tau_{yz}^{k},\tau_{xz}^{k}\right)} = \overrightarrow{\left(\tau_{xy}^{k+1}-\tau_{xy}^{k},\tau_{yz}^{k+1}-\tau_{yz}^{k},\tau_{xz}^{k+1}-\tau_{xz}^{k}\right)} \end{cases} \tag{10}$$

The evaluation index of shear stress increment is further defined as:

$$f(\boldsymbol{\tau}) = |d\boldsymbol{\tau}| = \sqrt{\left(\tau_{xy}^{k+1}-\tau_{xy}^{k}\right)^2+\left(\tau_{yz}^{k+1}-\tau_{yz}^{k}\right)^2+\left(\tau_{xz}^{k+1}-\tau_{xz}^{k}\right)^2} \tag{11}$$

5.4.3. Shear stress index (τ_{ry}/I_1)

Because the shear stress is the direct cause of the principal stress rotation, and τ_{ry} is much larger than τ_{zy} and τ_{rz}, the shear stress indicator is reduced to:

$$\frac{|\boldsymbol{\tau}|}{I_1} = \frac{\sqrt{\tau_{ry}^2+\tau_{yz}^2+\tau_{zr}^2}}{I_1} \approx \frac{\tau_{ry}}{I_1} \tag{12}$$

Therefore, τ_{ry}/I_1 is used as the index for evaluating the degree of principal stress rotation. It can be defined as the disturbance degree of

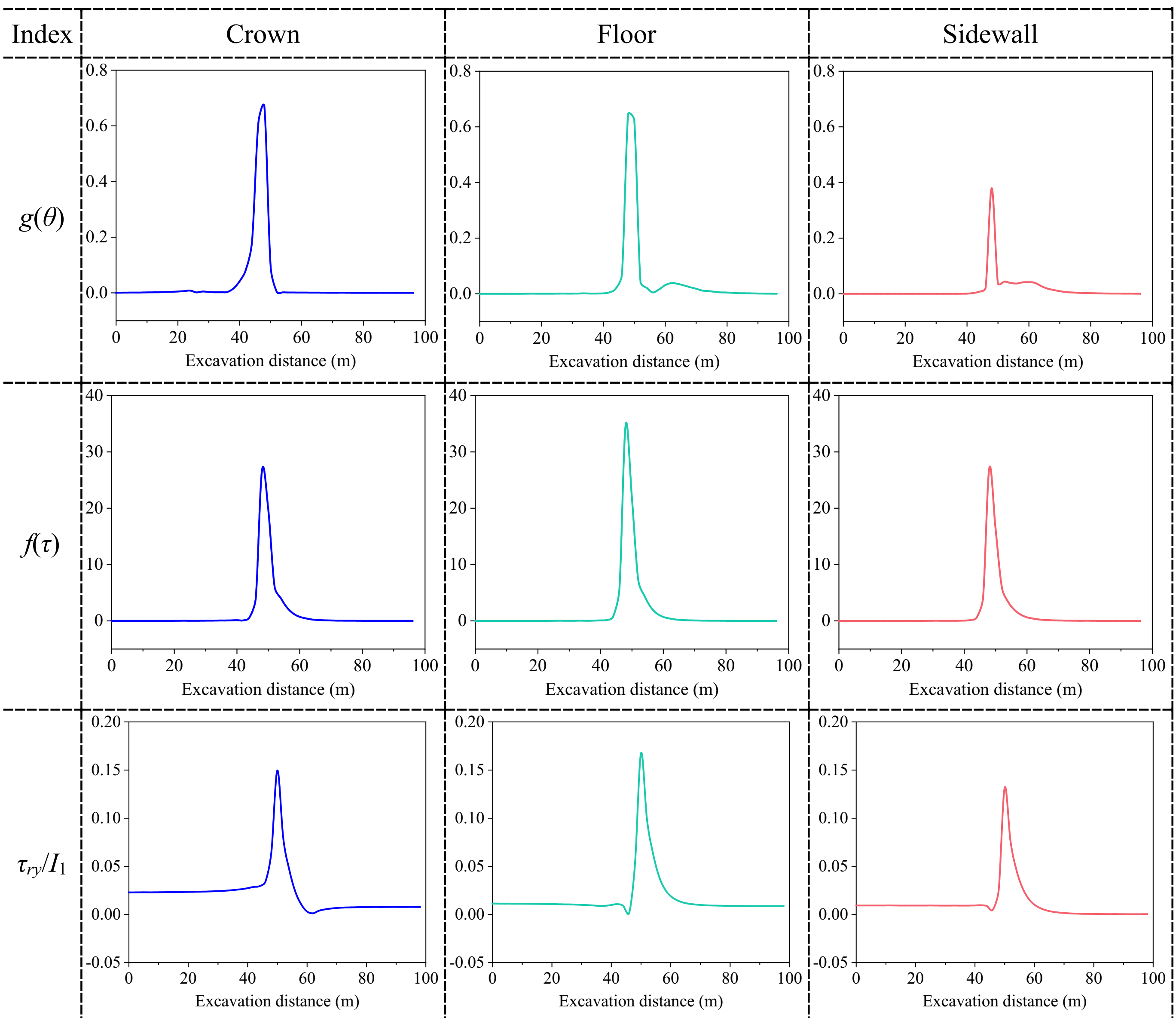


Fig. 11. Computed principal stress rotation under different evaluation indices (i.e., $g(\theta)$, $f(\tau)$, τ_{ry}/I_1).

shear stress (τ_{ry}) to the overall stress magnitude (I_1), and reflects the influence of shear stress on normal stress.

The principal stress rotations of the rock units at the tunnel crown, floor, and sidewall, evaluated using $g(\theta)$, $f(\tau)$, and τ_{ry}/I_1, are shown in Fig. 11.

The degree of principal stress rotation at the tunnel floor is slightly larger than that at the tunnel crown, while that at the tunnel sidewall is relatively small (Fig. 11). The curve characteristics between the different evaluation indicators ($g(\theta)$, $f(\tau)$, τ_{ry}/I_1) are consistent and can be used to evaluate the degree of principal stress rotation. However, $g(\theta)$ and $f(\tau)$ are incremental indicators between adjacent excavation steps, and there may exist large fluctuations due to the excavation method and cyclic footage; hence, it is not suitable to compare the degree of principal stress rotation when the tunnels adopt different excavation footages or construction methods.

This study uses τ_{ry}/I_1 to evaluate principal stress rotation for the following reasons: (1) τ_{ry}/I_1 is the state quantity and an absolute stress index that is only related to the excavation state and not affected by excavation processes, such as cycle footage; (2) τ_{ry} is the direct cause of the principal stress rotation, and superposition of normal stress in the plane strain coordinate system and τ_{ry} causes the rotation; (3) I_1 reflects the overall stress level and does not change with the selection of coordinate system for a rock unit, but varies at different buried depths; (4) the index τ_{ry}/I_1 is dimensionless and can accurately reflect the perturbation effect of shear stress on the normal stress (or stress level) and can also be used to evaluate the principal stress rotations of the tunnels under different buried depths.

5.5. *Comparison of rotation degree of tunnels under different buried depths*

Fig. 12a-c shows the quantitative results of the principal stress rotation degrees using τ_{ry}/I_1 for tunnels with different buried depths, which are consistent with the results obtained in Fig. 11. The principal stress rotation of the tunnel floor is higher than that of the tunnel crown and sidewall. The degrees of rotation for the tunnels with different buried depths is slightly above zero at distances >5 m away from the excavation face, but changes drastically within 5 m from the face. Furthermore, the degree of principal stress rotation gradually increases as the buried depth increases, and τ_{ry}/I_1, or rotation, is more prominent in deep tunnels, thus indicating that deep tunneling stability is largely affected by stress rotation.

Section 5.2 explains that the rock damage or surrounding rock collapse caused by principal stress rotation under different stress magnitudes can vary. The rotation degree in the deep tunnel is much greater than that in the shallow tunnel (Fig. 12), and the potential risk of rock instability caused by stress rotation in the deep tunnel is much more severe than that in the shallow tunnel; τ_{ry}/I_1 increases by 30% – 50% more for the 2000 m tunnel compared with the 400 m tunnel. The high principal stress rotation in the deep tunnel will further cause instability and failure of the tunnel rock.

6. Advanced core rock strengthening and stress control for surrounding rock stability

Sections 4 and 5 show that the 3D extrusion effect and complex stress path of surrounding rock elements near the tunnel face are significant in deep tunnels, and are the decisive factors for the stability and construction safety of deep tunnels. Therefore, it is important to address the potential engineering catastrophe induced by the 3D space effect near the tunnel face under high ground stress in the control and design of deep tunnels. In this section, the strengthening effects of core rock parameters are investigated.

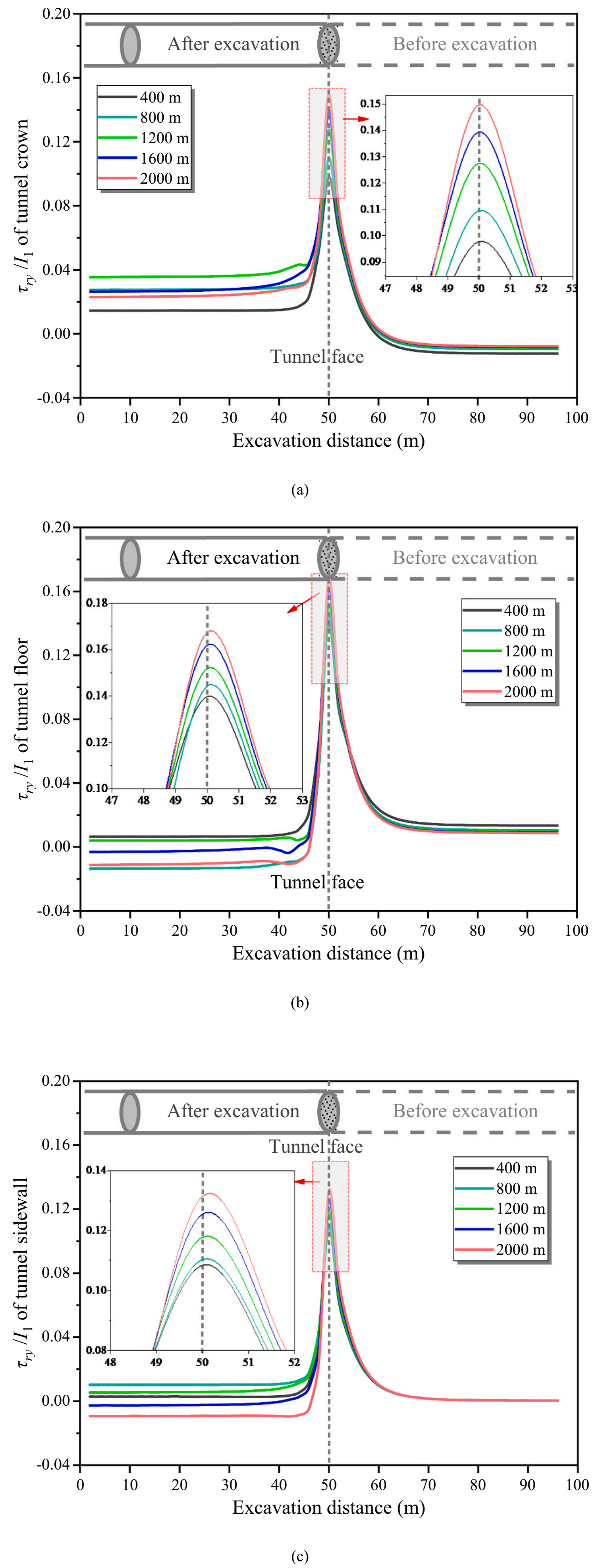


Fig. 12. Comparison of principal stress rotation degree for tunnels under different buried depths: (a) crown, (b) floor, and (c) sidewall.

6.1. Core rock mass strengthening based on GSI

6.1.1. Strengthening range

A significant 3D space effect exists within the range of 2–5 m from the tunnel face (see Section 5.3). Therefore, the strengthening ranges of the core rock are set as 2, 4, and 6 m.

6.1.2. Strengthening parameters

GSI is a comprehensive parameter that reflects the structural information relating to the mechanics and strength properties of the rock mass. The *GSI* is set to 46 (original rock), 51, 56, 61, and 66 in each numerical modeling to explore the stability control effect and principal stress rotation of the rock under different *GSI*. The corresponding GZZ criterion parameters (m_b, s, a) can be calculated using Eq. (5). In addition, the rock mass elastic modulus and compressive strength can be calculated using Eqs. (13) and (14).

$$E_{mk} = \left[\frac{1 + e^{(75+25D-GSI_0)/11}}{1 + e^{(75+25D-GSI)/11}}\right]E_{m0} \tag{13}$$

$$\sigma_{ck} = \frac{c_k cos\varphi_k \cdot (1 - sin\varphi_0)}{c_0 cos\varphi_0 \cdot (1 - sin\varphi_k)}\sigma_{c0} \tag{14}$$

where $k = 0, 1, 2, 3, 4$, corresponding to *GSIs* 46, 51, 56, 61, and 66, respectively. E_{mk} is the elastic modulus of rock mass; E_{m0} is the original elastic modulus, obtained from the geological survey data and set at 6270 MPa; $\sigma_{c0}$is the original UCS (140 MPa); c_k is rock mass cohesion, obtained from m_b, s, a (Hoek et al., 2002).

The strengthening parameters under different *GSIs* are tabulated in Table 2:

6.2. Comparison of stress states of core rock before and after strengthening

As shown in Fig. 13, the stress level of the core rock element gradually increases as the *GSI* increases. The radius of the strength envelope (GZZ) also gradually increases on the π-plane. In addition, the rock stress point gradually shifts from the outer to inner strength envelope and changes from plastic to elastic stress state. The σ_3 of the unstrengthened rock element is negative, indicating a volatile tensile stress state ($\sigma_3 < 0$) that transitions into the compression state ($\sigma_3 > 0$). The gradual increase of σ_3 along with the active control method described in Section 5.1 significantly improves the rock strength and stability.

However, the overall differences (strength envelope radius, rock stress state, and the relative relationship between them) among the different strength ranges (i.e., L = 2, 4, and 6 m) under the same *GSI* are insignificant, thereby indicating that the effect of the strengthening parameter on the rock stability is more significant than the strengthening range.

6.3. Extrusion deformation of tunnel face before and after strengthening

As shown in Fig. 14, the extrusion deformation of the tunnel face initially decreases rapidly and then gradually stabilizes with increasing *GSI*. The extrusion is sensitive to changes in *GSI*. A low *GSI* range shows a significant strengthening effect on the inhibition of tunnel face extrusion for soft rock (poor quality with low *GSI*) and limited effect for hard rock (good quality with high *GSI*). The difference of extrusion deformation under different reinforcement ranges (L = 2, 4, 6 m) is very small; similar conclusions can also be obtained for the *YAI* (Fig. 14). The rock element of the tunnel face center is transformed from the plastic to elastic stress state when GSI = 61–66 according to the *YAI* curve, agreeing with the results in Fig. 13. This suggests that GSI = 61–66 and L = 4 m could be taken as the economic strengthening parameters and range, respectively.

Table 2
Strength parameters of core rock mass in the deep tunnel under varied *GSIs*.

GSI	σ_{ci} (MPa)	m_b	s	a	c (MPa)	φ (°)	E_m (GPa)	ν_m
46	140	1.06	0.00112	0.508	3.93	32	6.27	0.28
51	161	1.32	0.00211	0.505	4.34	33	9.70	0.28
56	185	1.64	0.00395	0.504	4.81	35	14.87	0.28
61	214	2.03	0.00741	0.503	5.36	37	22.47	0.28
66	250	2.52	0.01389	0.502	6.03	39	33.25	0.28

6.4. Relationships of re-extrusion and pre-convergent deformation of core rock mass

Fig. 15 shows the relationships between pre-extrusion of core rock and pre-convergence deformation of core surrounding rock for the tunnel crown (Fig. 15a), tunnel floor (Fig. 15b), and tunnel sidewall (Fig. 15c). The red dotted lines in Fig. 15 are the connecting lines of the tunnel face sections (the end points of each curve) under varied *GSIs*. The pre-extrusion and pre-convergence deformations are both controlled by gradually increasing the *GSI*. The red dotted line gradually approaches the origin point, and the end of each curve passes through the origin point. Based on these results, it is assumed that both the face extrusion and pre-extrusion deformation are close to zero if the core rock has enough strength and rigidity.

Pre-extrusion and pre-convergence deformation are positively correlated, as shown in Fig. 15. The differences in the relationship curves under different *L* is insignificant, showing the limited effect of strengthening ranges. However, different *GSIs* correspond to different curve partition ranges; the larger value of *GSI* corresponds to the shorter curve, thereby indicating that strengthening affects the pre-extrusion and pre-convergence deformation of the core rock. The curve trends under different *GSIs* have relatively small differences, with each cluster of curves following a line. The relationship between pre-extrusion and pre-convergence of core rock follows a consistency law, which is given by:

$$f(u_y, u_r) = 0 \tag{15}$$

The consistency law suggests that the pre-convergence deformation of core surrounding rock can only be controlled if the extrusion (or pre-extrusion) deformation of the core rock is controlled, regardless of the construction method (New Austrian or ADECO-RS methods); the strengthening parameters and range are the only factors inhibiting the extrusion deformation.

6.5. Strengthening effect on the principal stress rotation

The principal stress rotation during excavation poses a considerable threat to the stability and safety of a deep tunnel. The τ_{ry}/I_1 index under different *GSIs* is used to evaluate the strengthening effect on the principal stress rotation degree. Fig. 16 describes the longitudinal distribution of τ_{ry}/I_1 under various *GSIs* for L = 4.

The longitudinal τ_{ry}/I_1 curves corresponding to the *GSIs* have similar curve shapes. τ_{ry}/I_1 increases sharply near the tunnel face, where the degree of principal stress rotation increases rapidly. The rotation degrees of the rock at the tunnel floor are the most severe, followed by the tunnel crown and sidewall. The τ_{ry}/I_1 under *GSI* = 66 is reduced by 20% − 30% compared with the unreinforced rock mass, therefore indicating that the strengthening of core rock restrains the stress rotation during the excavation to a certain extent.

Though the surrounding rock has not been strengthened in this study, the principal stress rotation of surrounding rock can be effectively suppressed by strengthening the core rock. Simultaneously, the negative effects induced by stress rotation, such as decrease in rock strength, weakening of rock mass parameters, and change in failure mode, are also reduced. However, close relationships among the pre-extrusion

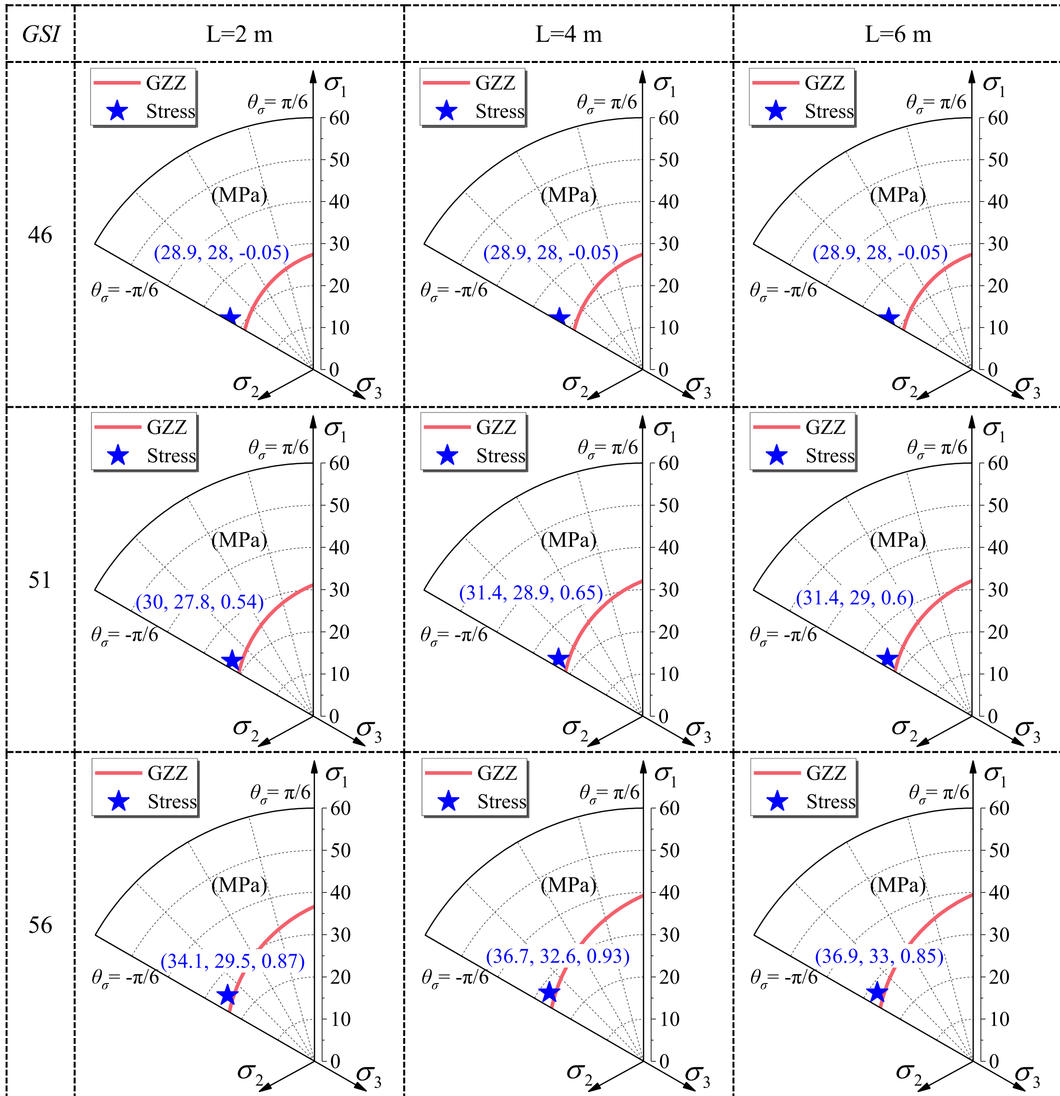


Fig. 13. Stress states of core rock under different strength range and *GSI*.

deformation of core rock, pre-convergent deformation of core surrounding rock, and convergent deformation of surrounding rock reflect the importance of the core rock mass on the overall stability of tunnel rock near the tunnel face. Hence, the remarkable 3D space effect makes it difficult to directly transfer the design and construction methods used in shallow tunnels to deep tunnels.

7. Discussion

The 3D space effect near the tunnel face and the stability of the surrounding rock was found to be related to the core rock strength and less affected by the reinforcement range. In addition, the pre-convergent deformation of the core surrounding rock was caused by the extrusion deformation at the tunnel face. Furthermore, the consistency law states that the pre-extrusion-pre-convergence of the core rock is minimally affected by the strengthening effect. Hence, the pre-convergence deformation and stability of the core surrounding rock is only affected by the extrusion deformation of core rock. Therefore, strengthening the core rock reduces the extrusion deformation and achieves indirect control of the pre-convergence deformation. The commonly applied New Austrian method controls the pre-convergent deformation of the core surrounding rock by directly constraining the extrusion deformation at the tunnel face. In contrast, the ADECO-RS method controls the pre-convergent deformation by increasing the strength of core rock to control the pre-convergent deformation indirectly.

As shown in Fig. 17, the extrusion deformation control of the tunnel face (e.g., New Austrian method) and strengthening of the core rock mass (e.g., ADECO-RS methods) both reduce the extrusion deformation at the tunnel face, maintain the stress inside the core rock, and reduce the stress unloading effect during excavation. In addition, σ_3 gradually increases and transforms from tensile to compressive stress. The nonlinear characteristics of rock mass strength show its sensitivity to changes in σ_3 at low stress levels. The mechanical behavior of the strengthened core rock mass (point *B* in Fig. 17) is enhanced significantly more than the unstrengthened rock (point *A* in Fig. 17). σ_1 of core rock is simultaneously severed with the σ_3 in the core surrounding rock, and the strength of core surrounding rock is further increased to control the pre-convergent deformation. Hence, the mechanical path of deformation control of the core rock is summarized as core rock (pre-extrusion deformation), core surrounding rock (pre-convergent deformation), and surrounding rock (convergent deformation near the tunnel face).

Stress unloading is more likely to cause rock damage and collapse than stress loading. Therefore, reducing the unloading effect (increasing σ_3) through active support methods can better control rock stability than

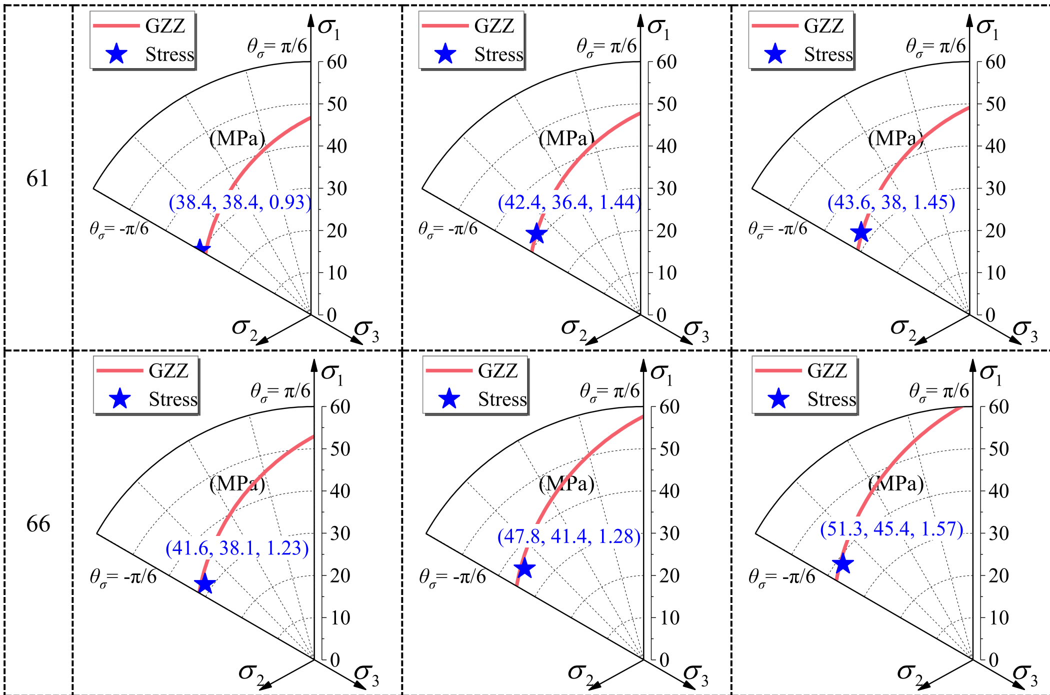


Fig. 13. (*continued*).

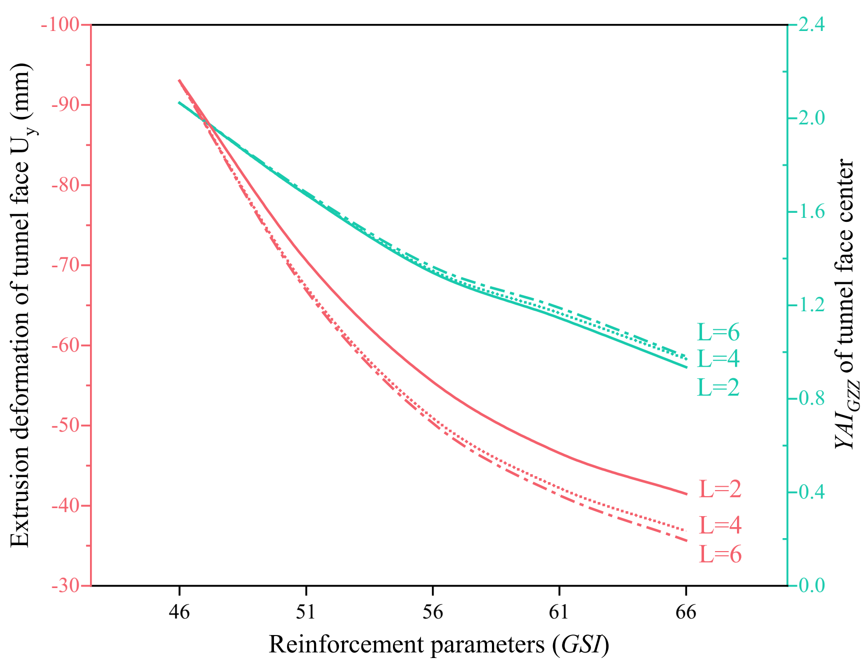


Fig. 14. Extrusion deformation and *YAI* of the core rock.

passive support (reducing or shifting σ_1) (Li, 2003; Li et al., 2017; Feng et al., 2020; Wang et al., 2020). However, it is believed that this conclusion is not exactly suitable for tunnel support design. As shown in Fig. 18a, it is relatively easy to adjust the rock stress within the strength envelope (elastic stress state) through active control methods (e.g., New Austrian method or ADECO-RS methods) when σ_3 is very low for core rock mass near the tunnel face. However, for the high stress state, shown in Fig. 18b, it is difficult to achieve the elastic stress state by increasing σ_3, as the active control method shows poor stabilizing effect of the tunnel rock. Reducing σ_1 through passive support offers better and effective results in this case. However, in most cases, the rock stress is in the state shown in Fig. 18c, and it may be better to find an optimal stress control path according to the relationship between stress state and strength characteristics by simultaneously increasing σ_3 and reducing σ_1.

The active control method of restraining core rock extrusion deformation can better stabilize the tunnel rock by effectively suppressing the stress unloading effect caused by excavation and reducing the rotation degree of principal stress. The stress magnitude in deep tunnel excavation is widely distributed, from zero at the surrounding rock to in situ stress far away from the tunnel. The optimized support methods should be selected according to the rock stress state and strength characteristics

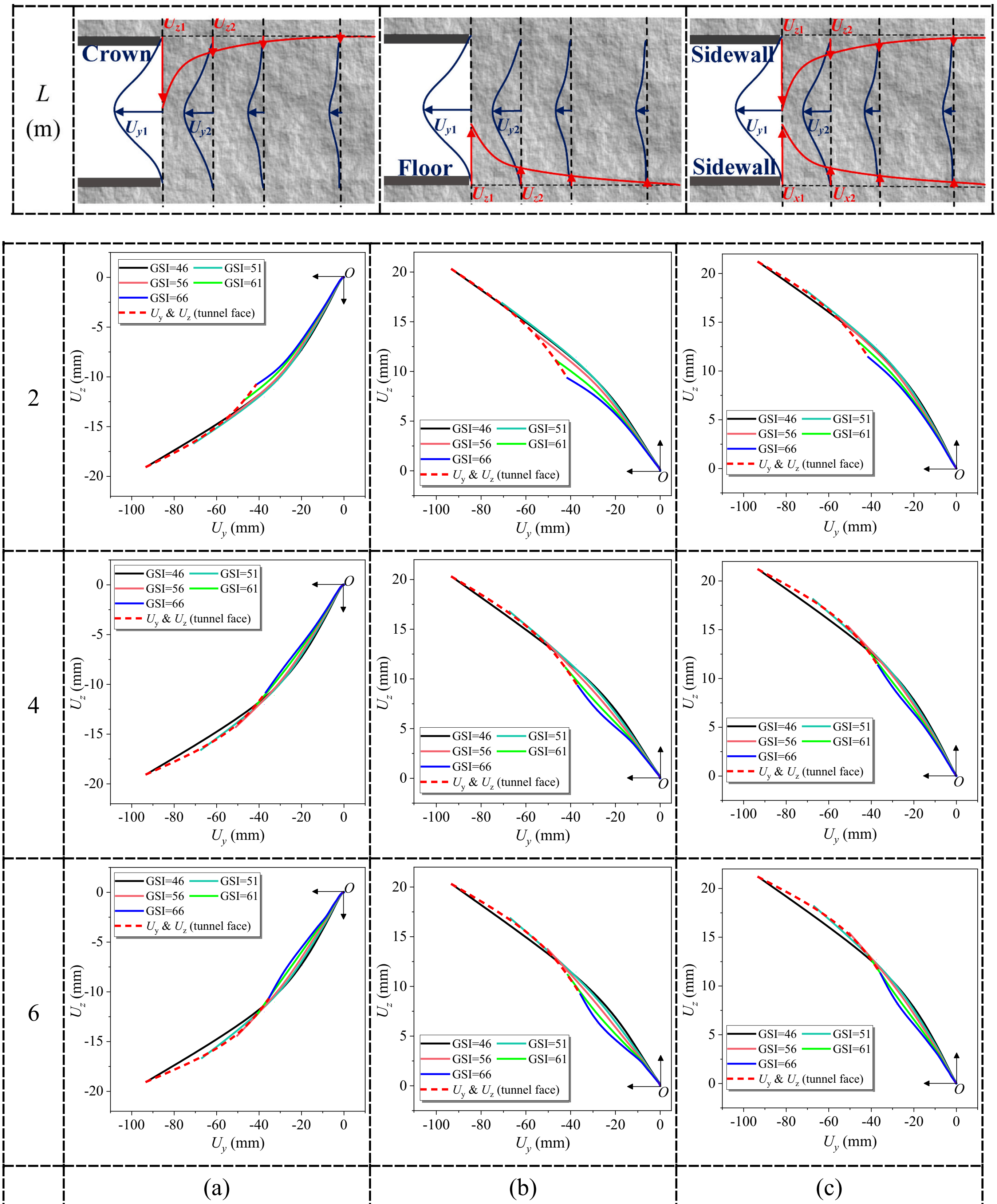


Fig. 15. Relationships of pre-extrusion (U_y) of core rock and pre-convergent deformation (U_r) of core surrounding rock for (a) tunnel crown, (b) tunnel floor, and (c) tunnel sidewall.

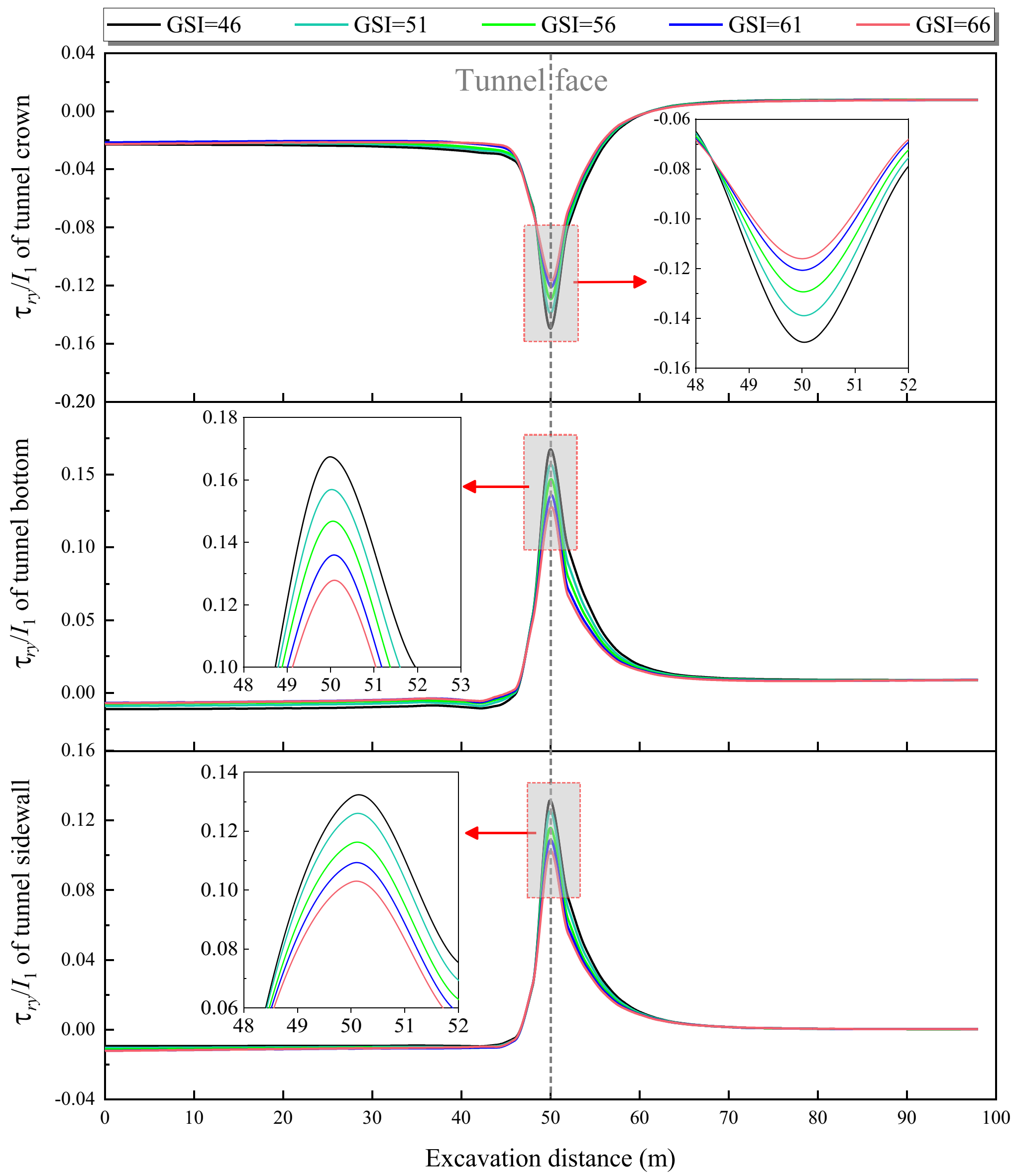


Fig. 16. Principal stress rotation under different *GSIs* (L = 4).

of the tunnel rock, i.e., active-passive methods in Fig. 18c, to ensure tunnel rock stability and construction safety near the tunnel face and to reduce the potential of a tunnel catastrophe caused by the significant 3D space effect and complex stress path in the operation of a deep tunnel. This will be further discussed and researched in the near future.

8. Conclusions

This study conducted a forward numerical analysis based on the smoothened GZZ strength criterion to analyze the 3D space effect, principal stress rotation, and stress control of the core rock during deep tunneling. The conclusions are as follows:

(1) The stress unloading and face extrusion of deep tunnel excavation are more significant than those of shallow tunnels. The σ_3 of the rock at the tunnel face gradually decreases with increasing buried depth and eventually becomes tensile. The plasticity or failure zone is initially generated at the tunnel face, and it is less stable compared with surrounding rock.

(2) The surrounding rock experiences significant stress rotations as the tunnel face approaches and passes through the rock element because of the sharply increased shear stress (τ_{ry}) near the tunnel face. The value of τ_{ry}/I_1 was thus developed and suggested to quantitatively evaluate the principal stress rotation during excavation. τ_{ry}/I_1 changed drastically near the tunnel face, and its value in the deep tunnel was much higher than that in the shallow one, indicating the potential risk of rock instability caused by stress rotation in deep tunneling.

(3) Strengthening of core rock can greatly improve the stability and mechanical behavior of tunnel rock, increasing the rock strength and reducing the principal stress rotation (τ_{ry}/I_1 decreased by 20% − 30% compared with the unreinforced one). The effect of the strengthening parameter, *GSI*, on rock stability was stronger than that of the strengthening range, and *GSI* can thus be selected as the main controlling factor in stability control of the tunnel face. The relation of pre-extrusion and pre-convergence deformation of core rock was discovered to follow a good consistency law, and the pre-convergence deformation of core surrounding rocks could only be affected by the extrusion of core rock.

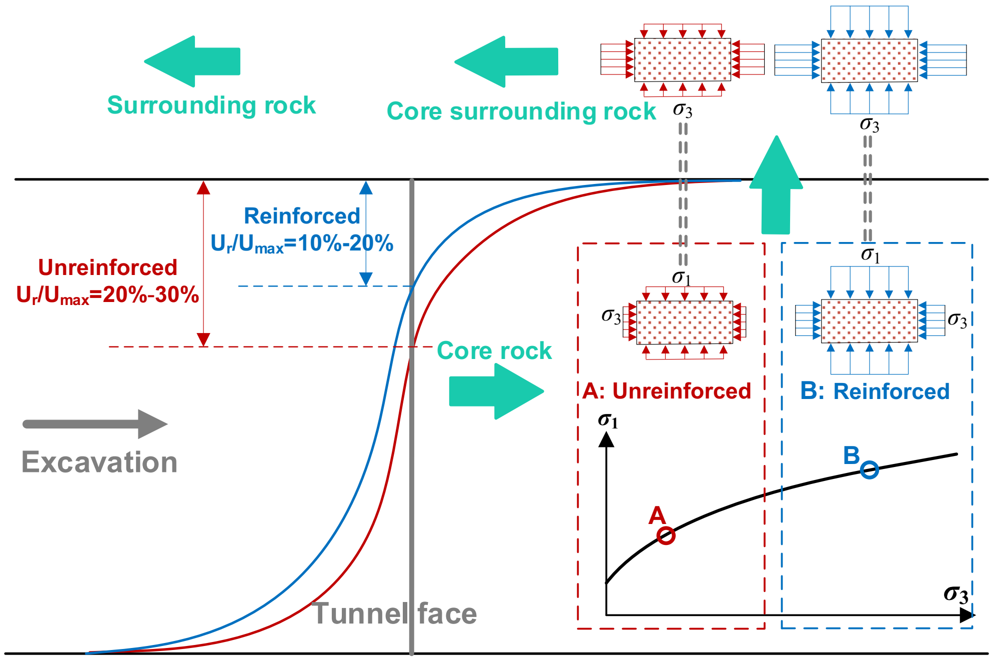


Fig. 17. Deformation control mechanism of core rock mass.

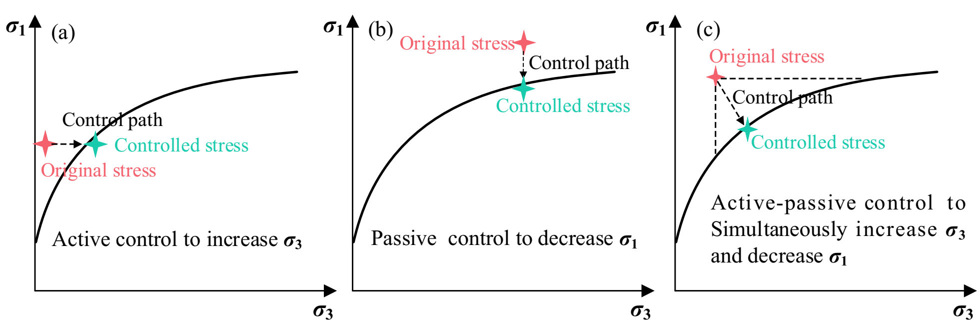


Fig. 18. Active-passive control mechanism using strength behavior of rock mass.

(4) The stress control of core rock is as follows: advanced core rock (pre-extrusion)–core surrounding rock (pre-convergent)–surrounding rock (convergent), with close relationships among them. The stability control of deep tunneling should focus on the extrusion of the tunnel face and principal stress rotation during excavation. The tunnel face control-based active-passive method could weaken the 3D space effect and reduce stress rotation, which would activate the strength potential and self-bearing nature of the rock, and may play an important role in the design and construction of deep and ultra-deep tunnels (e.g. Sichuan-Tibet Railway).

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CRediT authorship contribution statement

Wuqiang Cai: Investigation, Methodology, Software, Data curation, Formal analysis, Writing – original draft. **Hehua Zhu:** Supervision, Conceptualization, Project administration, Resources. **Wenhao Liang:** Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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